
System1Bench: Typed Decisions Across Domains and Matched Interventions

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Abstract

Typed decision interfaces turn a supplied state into finite choices that an application can act upon, yet a single accuracy score hides both source heterogeneity and sensitivity to equivalent presentations. System1Bench evaluates five fixed model–adapter systems across 15 source collections (28 main suites and eight matched controls), then tests independently sourced legal and scientific evidence pairs. We preserve domain, decision function and reference provenance, and report paired correctness, prediction transitions, invalid outputs and efficiency instead of pooling incompatible labels. On 144 balanced ContractNLI test pairs, candidate reversal switches 66 Llama and 64 Qwen labels, even as their reference accuracies rise by 10.4 and 12.5 percentage points. In a post-hoc census of all 339 eligible SciFact cited pairs, Qwen’s total reference-correct count changes only $231 \rightarrow 220$, while no-information correct falls $112 \rightarrow 68/130$; a fresh 159-pair same-source complement to an earlier balanced selection preserves that class-level loss. A 2×2 legal control shows that the same apparent reversal hides different adapter couplings: answer-code remapping alone switches 125/144 Llama labels but 10/144 Qwen labels; display reversal alone switches 6/144 and 36/144. Exact-score ties require an explicit common decoding rule for interpreting that control. A document-blind, supervised hypothesis prior reaches 56.9% on the selected legal cases, underscoring source-shortcut risk. The artifact retains source-level denominators, raw predictions and executable references. These findings characterize typed downstream decisions, not end-to-end system competence.

1 Introduction

A decision interface can appear reliable under aggregate accuracy while changing its answers when the same candidate set is presented in a different order. This discrepancy matters when an application routes a request, triggers a review, or selects an action from a finite schema. The application consumes the selected action, rather than the model’s average score over unrelated examples. Evaluation must therefore ask both whether a decision matches its reference and whether equivalent presentations preserve that decision.

System1Bench studies interfaces that map a supplied state and typed questions to structured answers. The motivating systems include Laya and Jev, alongside language models constrained to the same answer spaces [Convai Innovations, 2026, TypeSafe AI, 2026]. The term “System 1” identifies this product/interface category. Our unit of comparison is a *model plus adapter on a fixed downstream decision*, not an entire deployed workflow. Whole-system success also depends on retrieval, tools, human review and action execution, none of which we measure.

The motivating empirical result is concrete. On the same 100 Banking77 control states, reversing candidate presentation changes 48% of Laya English predictions while increasing reference accuracy by nine percentage points. Laya Multilingual changes 38% of predictions with zero net accuracy change. A leaderboard displaying only the two marginal accuracies would omit most of the behavioral

change. The comparison is not a universal degradation claim: it identifies sensitivity that has no fixed direction in aggregate correctness.

Independent natural-domain references sharpen this observation and expose its limits. We froze selected ContractNLI test document–hypothesis pairs and SciFact development claim–abstract pairs before the new runs, retaining their distinct three-way and positive-evidence label spaces rather than treating either as generic classification. A later post-hoc same-source census covers all 339 eligible SciFact cited pairs and tests sensitivity to the earlier balanced three-class selection; it is not an independent holdout. A candidate reversal changes dozens of legal decisions for the two 8B code-likelihood adapters, but its net accuracy direction varies across source–task–adapter configurations. More importantly, a four-cell legal control shows that a seemingly simple reversal changes two separable interface features—display position and the semantic label assigned to an answer token—and that the models respond to those features in sharply different ways. This turns a stability symptom into a testable adapter-level diagnosis, without claiming a causal change in legal reasoning.

Our approach combines a source atlas with matched perturbation diagnostics. Sources differ in application domain, decision function, reference provenance, output type, candidate cardinality, language, context length and the number of questions sharing a state. We preserve these distinctions instead of averaging unlike references. Paired tests compare the same semantic unit across encodings; same-order repeats provide an observed variability reference, not by themselves a causal estimate of a presentation effect. Separate operational experiments measure resident-model completion and hosted calls under their respective observable boundaries.

This paper contributes an auditable *evaluation design*, rather than claiming that perturbation sensitivity or multidimensional scoring is new. First, it instantiates one typed interface across 15 atlas collections, then adds independently sourced legal and scientific diagnostics, while retaining sample identity, non-exclusive task functions and heterogeneous reference provenance; source-level results are not converted into a blended accuracy ranking. Second, it applies paired correctness, turnover and prediction-discordance accounting to native decision heads and code-scored LLM adapters, and uses an orthogonal four-cell control to separate candidate position from answer-code mapping. An executable-policy follow-up and a held-out wording test constrain a post-hoc explanation of another striking result. Third, it supplies source-shortcut audits and reproducible quality and efficiency records with distinct resident-GPU and hosted-API timing boundaries. The historical analyses are exploratory: their outputs existed before the diagnostic questions were finalized; each generated or sourced follow-up records when its inputs and conditions were fixed relative to inference.

The central implication is methodological. Stable accuracy, stable decisions and efficient execution are distinct properties. A practical benchmark should reveal which of them a system satisfies on a particular workload.

2 Related work

Behavioral and multidimensional evaluation. Behavioral testing organizes model evaluation around capabilities and controlled transformations rather than a single held-out score. CheckList formalizes minimum-functionality, invariance and directional-expectation tests for NLP systems [Ribeiro et al., 2020]. System1Bench adopts this separation at a finite-decision interface: candidate order, language realization and irrelevant state fields define invariance-style comparisons, while an operative fact change defines a directional test with recomputed executable labels. The relevant unit is a shared state cluster, not every transformed row treated as independent evidence.

HELM broadens evaluation across scenarios and metrics, emphasizing that accuracy alone leaves substantial behavior unmeasured [Liang et al., 2022]. Our setting is narrower: finite typed decisions under explicit candidate and rubric contracts. This narrower interface enables exact prediction matching and shared-state perturbations across native decision heads and code-scored language models. It also requires preserving distinctions between dataset annotations, authored policy references, synthetic teachers and programmatic labels. A large count of decisions cannot substitute for an explicit account of what those references establish.

Recent decision-model studies make a stronger comparison necessary. Legal-document evaluation of Jev and language models already shows that equal aggregate scores can mask different persistent decisions, corrections and regressions [Zhang et al., 2026]. Computational-social-science text-

annotation work evaluates 18 tasks and many language and decision systems with quality, calibration, routing and cost criteria [Ibrahim and Zaki, 2026]. JevAdvBench evaluates Jev’s typed interface across vendor-derived scenarios using exact re-runs, single-edit attacks and confidence-gate outcomes [Hu et al., 2026]. Its primary flip metric deliberately avoids treating model-derived consensus labels as independent gold. System1Bench is not the first cross-domain or stability study. Its narrower complementary question is how correctness transitions and representation sensitivity compare *across model–adapter systems* on fixed downstream decisions with source-specific references, including executable rules where correctness is determined by visible facts. Public labels, authored cases and synthetic teachers remain separate evidence tracks rather than interchangeable ground truth.

Rule-governed decision benchmarks also evaluate whether models apply external rules to facts; RGDt-Bench additionally measures the completeness of stated justifications [Zhao et al., 2026]. Our generated policies instead provide an executable oracle for paired answer interventions and do not evaluate explanations or natural-world outcomes. These studies motivate treating broad source coverage as an evaluation substrate, not a novelty claim by itself.

Finite-choice presentation. Prior work documents option-identifier bias and benchmark sensitivity to choice order and answer-selection method [Zheng et al., 2023, Alzahrani et al., 2024]. Candidate-order sensitivity and code bias are therefore established problems. In our fixed single-token LLM adapter, the codebook control separately changes semantic display order and label-to-code assignment, whereas ordinary reversal changes both. It identifies an interaction in that adapter, not a new general token-bias mechanism or an architectural property. Moreover, first-token scoring can diverge from a model’s generated textual answer [Wang et al., 2024]; our LLM results characterize a constrained decision interface rather than unconstrained generative capability.

Cross-lingual and contextual behavior. XNLI supplies cross-lingual inference examples with shared labels, supporting paired evaluation of language realizations [Conneau et al., 2018]. We use the source alignment to compare English and Chinese under the existing question wrapper. Translation realization and wrapper language are part of this intervention; the paired result describes the implemented interface. CLINC150 similarly separates in-scope intent classification from explicit out-of-scope prediction [Larson et al., 2019]. That distinction is operationally relevant because a finite choice among known actions does not by itself supply a reliable reject option.

Long-context results depend on the position of the relevant information as well as the nominal context budget. Lost in the Middle demonstrates this dependence in long-context language-model tasks [Liu et al., 2024]. Our needle diagnostic retains source length and insertion position, but also separates the finite-choice intent question from a binary damage question attached to the same state. Grouping by the original needle preserves dependence among length and position variants. The diagnostic concerns explicit fact retrieval, rather than complete long-horizon decision competence.

Selective prediction. Selective classification evaluates the trade-off between making predictions and rejecting uncertain cases [Geifman and El-Yaniv, 2017]. A useful finite-decision model should ideally concentrate its errors among low-confidence cases. We plot empirical risk against coverage within each source, accepting tied confidence values together. The common ranking statistic is maximum class probability. Native decision confidence and a language model’s softmax over candidate codes need not share calibration semantics, so selective curves characterize observed error ranking under each adapter rather than interchangeable confidence thresholds.

Inference measurement. MLPerf Inference distinguishes operating scenarios and requires performance to be considered with quality constraints [Reddi et al., 2020]. PagedAttention evaluates serving efficiency under an explicitly specified workload and system configuration [Kwon et al., 2023]; Sarathi-Serve studies the throughput–latency trade-off in LLM inference [Agrawal et al., 2024]. We use these principles to declare timing boundaries, workload shapes, hardware, warm-up and replication. Our local experiment measures an uncached state-to-structured-answer call in a resident process. It excludes model loading, network transport and queueing. Hosted Jev calls expose a different boundary and remain a separate track. This separation is necessary for meaningful interpretation of model-adapter timing rather than a mixed comparison of service and local execution.

3 Task taxonomy and measurement design

3.1 Typed contract and reference provenance

An input comprises a state s_i and a question map Q_i . Each question specifies an instruction, an output primitive and, when applicable, a candidate set or ordinal rubric. A *choice* selects a finite label, a *noul* returns a binary probability, and a *score* returns a distribution over ordered levels together with its expectation. Model-facing payloads contain only state and questions; reference labels and hidden source metadata are excluded by an explicit whitelist.

The historical collection contains 15 source collections, instantiated as 28 main suites and eight repeat/reversal controls. Figure 1 maps each source to an application domain and non-exclusive decision functions: semantic classification, routing, evidence use, ordinal judgment, safety, workflow, explicit rejection and retrieval. These functions are descriptive tags, not equal-weight components of a latent intelligence score or independently replicated domain strata. A suite additionally records primitive, language, candidate count, state/question multiplicity and reference type. Main-suite variants can share underlying examples: BoolQ and SST5, for example, each have two output encodings.

References are classified as dataset-provided, human prompt annotations, authored or AI-reviewed policy cases, synthetic-teacher outputs, or programmatic labels. These classes describe provenance, not a common standard of truth: agreement with a synthetic teacher is teacher agreement, while policy labels express conformity to the supplied rubric. Source labels may be ambiguous or conflicting, so claims of noise-free correctness are restricted to defensible reference tracks. The source manifest pins downloaded artifacts and preparation code; request fingerprints preserve dictionary insertion order because candidate order is an experimental variable. Original model outputs and input audits remain immutable.

3.2 Correctness, discordance and paired effects

Let $\hat{y}_i^{(a)}$ be the canonicalized finite prediction under condition a , and let y_i be the reference. Invalid responses count as incorrect. Define $c_i^{(a)} = \mathbf{1}[\hat{y}_i^{(a)} = y_i]$. For a semantics-preserving intervention $a \rightarrow b$, we report

$$\Delta_{a \rightarrow b} = \frac{1}{n} \sum_i (c_i^{(b)} - c_i^{(a)}), \quad D_{a \rightarrow b} = \frac{1}{n} \sum_i \mathbf{1}[\hat{y}_i^{(a)} \neq \hat{y}_i^{(b)}]. \quad (1)$$

These quantities answer different questions. If N_{01} counts incorrect-to-correct changes and N_{10} correct-to-incorrect changes, then $\Delta = (N_{01} - N_{10})/n$. The *correctness turnover* $T = (N_{01} + N_{10})/n = |\Delta| + 2 \min(N_{01}, N_{10})/n$ exposes the opposing changes hidden by net accuracy. Discordance D also includes changed predictions that are both incorrect. We retain both transition counts, pair identifiers and source-cluster membership in the analysis artifact.

For candidate reversal, an original condition o , exact-repeat condition r and reversed condition v give the descriptive contrast $D_{r \rightarrow v} - D_{o \rightarrow r}$. It uses the same clusters in both terms, but subtracting an observed repeat rate does not alone identify a causal effect of order. It describes the implemented candidate presentation and answer-code mapping jointly. In the LLM adapter, changing candidate order also changes which answer code represents an option.

Uncertainty is estimated with 10,000 paired cluster-bootstrap draws and seed 20260929. The cluster is the source/state unit, including all questions and matched variants. Needle analyses resample the 25 original content units; their 450 derived states are not 450 independent samples. Intervals for historical contrasts are pointwise and descriptive. Complete contrast families, rather than only favorable directions, are retained.

3.3 Compared systems

The local systems are Laya English and Multilingual from the same pinned repository revision, Llama-3.1-8B-Instruct and Qwen3-8B. Laya uses native typed heads. The LLM adapter serializes each question and scores distinct single-token candidate codes with one forward pass, without generated reasoning. Its probabilities are conditional on those candidate logits. Supplied binary criteria and

Fifteen sources, eight decision functions

Historical suite atlas · 28 main suites · function tags overlap

SOURCE / DOMAIN	SEM	RTE	EVD	ORD	SFT	FLW	REJ	RET	REF
LANGUAGE & AFFECT									
AG News	●	●	●	●	●	●	●	●	D
News	●	●	●	●	●	●	●	●	D
Emotion	●	●	●	●	●	●	●	●	D
Social text	●	●	●	●	●	●	●	●	D
SST-5	●	●	●	●	●	●	●	●	D
Reviews	●	●	●	●	●	●	●	●	D
SERVICE & ROUTING									
Banking77	●	●	●	●	●	●	●	●	D
Banking support	●	●	●	●	●	●	●	●	D
MASSIVE	●	●	●	●	●	●	●	●	D
Assistant commands	●	●	●	●	●	●	●	●	D
CLINC150/OOS	●	●	●	●	●	●	●	●	D
Assistant intents	●	●	●	●	●	●	●	●	D
EVIDENCE & INFERENCE									
BoolQ	●	●	●	●	●	●	●	●	D
General QA	●	●	●	●	●	●	●	●	D
XNLI	●	●	●	●	●	●	●	●	D
Natural inference	●	●	●	●	●	●	●	●	D
TurtleBench	●	●	●	●	●	●	●	●	D
Puzzle evidence	●	●	●	●	●	●	●	●	D
SAFETY									
Prompt injections	●	●	●	●	●	●	●	●	D
Prompt security	●	●	●	●	●	●	●	●	D
Aegis2 prompts	●	●	●	●	●	●	●	●	H
Content safety	●	●	●	●	●	●	●	●	H
WORKFLOW FIXTURES									
Typed decisions	●	●	●	●	●	●	●	●	T
Mixed enterprise	●	●	●	●	●	●	●	●	A
JevBench	●	●	●	●	●	●	●	●	A
Mixed fixtures	●	●	●	●	●	●	●	●	A
ReflexBench	●	●	●	●	●	●	●	●	A
Product workflow	●	●	●	●	●	●	●	●	A
Jev-Laya	●	●	●	●	●	●	●	●	T/P
Mixed workflow	●	●	●	●	●	●	●	●	T/P

SEM semantic · RTE routing · EVD evidence · ORD ordinal
SFT safety · FLW workflow · REJ rejection · RET retrieval
REF: D dataset · H human · T teacher · A authored · P program

Figure 1: Cross-domain source atlas: 15 collections contributing 28 main suites, grouped by application domain. Filled cells mark non-exclusive decision functions. REF codes distinguish dataset labels (D), human prompt annotations (H), synthetic teachers (T), authored or AI-reviewed cases (A) and executable labels (P); they do not encode model accuracy or a common ground truth.

ordinal levels are preserved. All historical local decisions passed complete-input checks; checkpoint file hashes identify the actual local weights.

Hosted Jev. The pinned `jev-1.13.0` endpoint completed 22,934 request executions covering the same 26,450 decisions. 8 decisions failed the declared transport/response validity rule and remain in the accuracy denominator. The API reported 23,841,873 input tokens, corresponding to USD 1.001 at the retrieved public price; unavailable usage on failed calls is not included. Every returned version and answer is retained. Server tokenization and weights are not observable. Hosted client times include the proxy, network and service and are reported separately from resident-GPU measurements.

3.4 Fresh executable-policy diagnostic

The new diagnostic generates 288 unique base states across refund, access-control and routing policies using three fixed data seeds. Original action labels are stratified, and boolean/ordinal marginal distributions are reported rather than assumed balanced. Each policy includes a priority-ordered action, an escalation predicate and a severity rule. The correct outputs are computed directly from visible facts and policy parameters and checked by a separately expressed rules implementation.

Eight matched variants test exact repetition, action-order reversal, an authored Chinese rendering, irrelevant archived records, question paraphrases, equivalent choice encodings, and a one-field counterfactual, alongside the original. The counterfactual must change the correct action; all other variants preserve the reference labels. Generation rejects base states without such a one-field intervention. Thus the diagnostic is a curated compositional-rule distribution. The generator, data fingerprints, scoring contract and intervention scope are frozen before inference. Three generator seeds replicate data generation, not model training.

All five systems complete 6,912 decisions each in this diagnostic. The six predeclared contrasts per model/policy family are excess reversal discordance, Chinese/distractor/paraphrase action correctness effects, and choice-encoding effects on review and severity correctness. We use a max-standardized-deviation bootstrap with 10,000 draws stratified by the three data seeds, providing simultaneous intervals within each six-effect family, not across models or policies. Zero empirical variance is flagged rather than interpreted as population invariance. A five-percentage-point practical discussion threshold was fixed before inference; it is not a universal acceptance criterion.

Each LLM additionally answers 1,440 codebook-control decisions: baseline, exact repeat, display-only reversal, code-only reversal and both. Prompt tests verify that baseline matches the ordinary adapter and both matches its reversed-candidate rendering. The native decision interfaces do not expose an equivalent answer-code assignment. These additional contrasts are descriptive and use pointwise intervals.

3.5 Held-out wording challenge

A post-hoc inspection of the fresh routing results motivated a second, separately frozen experiment. We generated 96 new base states from seeds 404, 505 and 606, disjoint from the 288-state policy set, balancing the four outage/payment truth cells within each seed. Each state has English and Chinese *compact* and *explicit* policy renderings. Within a language, only the first policy clause changes; facts, parameters, questions, answer options and executable references are held fixed. The primary contrast, specified in the frozen protocol before inference on these states, is explicit-minus-compact action accuracy on the 48 exclusive-or states in English; Chinese and the remaining truth cells are reported as secondary. All five systems completed 384 requests each. Intervals use 10,000 paired bootstrap draws stratified by seed and truth cell. This is a controlled comparison of the tested authored clauses, not an independently adjudicated semantic-equivalence or translation study.

4 Representation sensitivity beyond aggregate accuracy

4.1 Candidate order exposes hidden instability

Figure 2 juxtaposes prediction flips and the opposing correctness transitions hidden by their net accuracy change. Across the four historical local systems and four control tasks, reversal flips range from 8.3% to 65.0%; same-order repeats range from zero to 5.0%. These measurements support a presentation-sensitivity finding, rather than a uniform accuracy penalty.

Laya English on Banking⁷⁷ illustrates the distinction: a 48% flip rate accompanies a nine-point accuracy increase. Laya Multilingual flips 38% of predictions with unchanged accuracy. Llama’s

Source	Model	Repeat flips	Reversal flips	Δ accuracy
banking77	Laya EN	0.0	48.0	+9.0
banking77	Laya Multi	0.0	38.0	+0.0
banking77	Llama	4.0	65.0	-25.0
banking77	Qwen	1.0	39.0	-18.0
banking77	Jev	1.0	3.0	+1.0
massive_en	Laya EN	0.0	50.0	-2.0
massive_en	Laya Multi	0.0	46.0	-7.0
massive_en	Llama	5.0	51.0	-2.0
massive_en	Qwen	0.0	32.0	-8.0
massive_en	Jev	2.0	10.0	-1.0

Table 1: Matched candidate controls. Flip rates and accuracy differences are in percentage points. Reversal is compared with the same-order repeat; the repeat flip rate uses the corresponding original predictions. Source sizes differ and are shown in Figure 2.

65% reversal flips exceed its four-percent repeat flips, while its accuracy falls by 25 points. A mean accuracy change neither determines the number of affected decisions nor separates presentation effects from execution variation. Recording the paired transition structure is therefore an evaluation requirement whenever stability matters to the application.

4.2 Output encodings and language realizations

The paired output-encoding effects depend on the task and system. On SST5, choice minus ordinal-score accuracy is +15.0, +6.1, +10.2 and -1.8 percentage points for Laya English, Laya Multilingual, Llama and Qwen respectively. BoolQ effects are smaller and differ in direction. These are comparisons of complete interface encodings: instructions, candidate descriptions and adapter execution can all differ. They identify interface sensitivity, not the causal effect of a particular neural head.

The English/Chinese comparisons share source labels and sample identities. Chinese-minus-English accuracy is negative for every historical local system on both XNLI and MASSIVE under the existing English question wrappers. The magnitude varies substantially by model and source. Figure 14 places paired correctness effects beside prediction disagreement; a smaller average drop can still accompany many changed decisions. The practical implication is to validate the actual multilingual request rendering used by an application rather than transferring an English threshold or score by name.

4.3 Context effects depend on the question

Figure 15 separates the choice and binary questions attached to each needle state. This split changes the interpretation of a pooled retrieval score. Llama retains strong choice retrieval across many lengths and positions while its binary damage question degrades with longer contexts. Laya English exhibits sharply different choice behavior when the relevant content appears at the start rather than the middle or end. Qwen’s curves are comparatively stable in this fixed diagnostic. These patterns argue for a length-by-position-by-question analysis rather than a single context-window number or an assumed monotonic degradation law.

4.4 Fresh policy interventions and codebook controls

Table 2 reports the original-state accuracies and joint original/counterfactual action success. The fresh diagnostic covers 96 base states per family, with eight matched variants. Figure 13 retains all six predeclared effects per model and policy, with simultaneous intervals within each six-effect family. The codebook experiment separates display order from answer-code identity for the two LLM adapters.

The fresh results separate invariance from supplied-rule competence. Jev achieves 100% original and joint action success on refund and access policies, but only 54.2% original and 29.2% joint success on routing. Small reversal effects on that policy therefore do not establish correct rule application. Several local model/policy cells are near the action-majority baseline; low intervention effects in

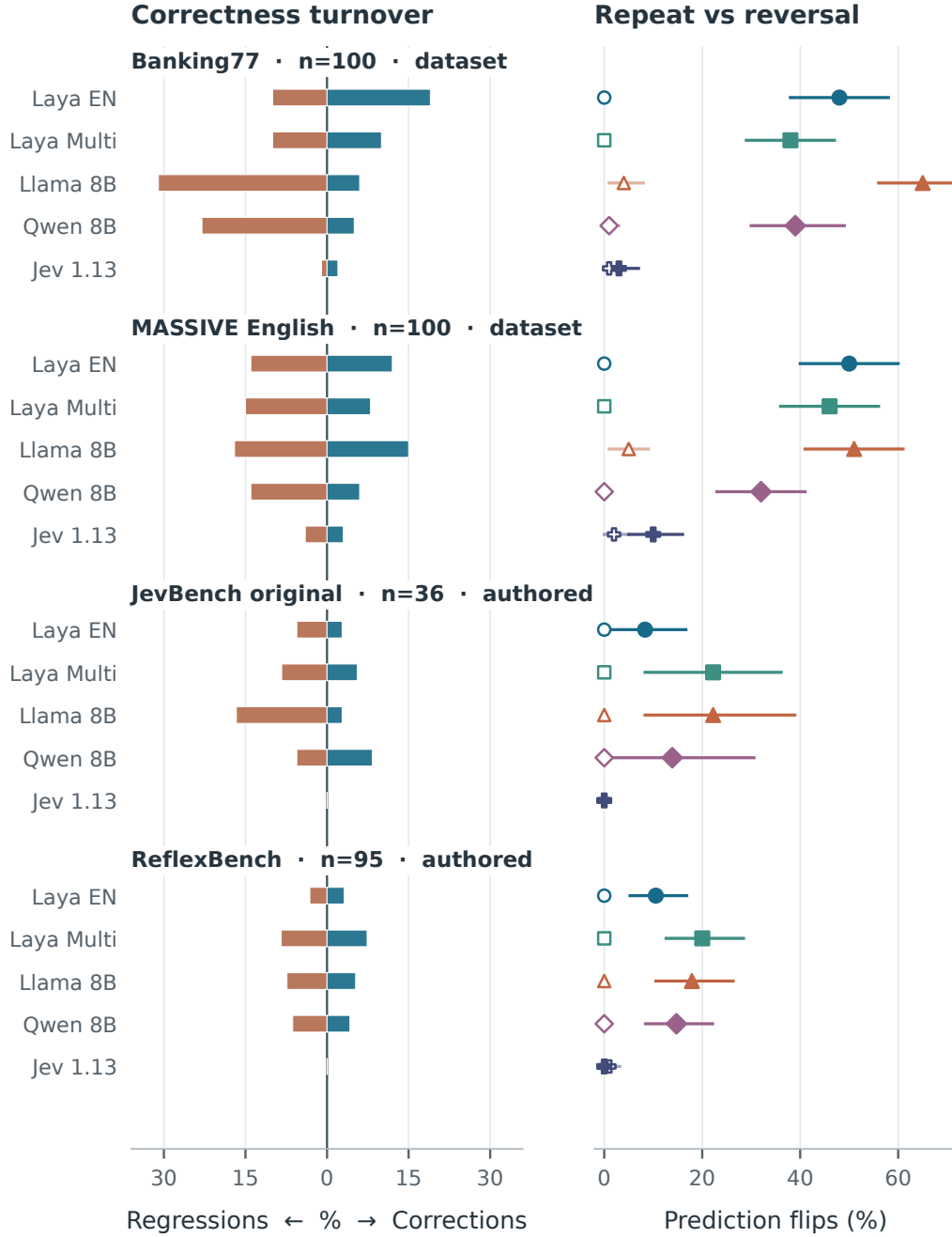


Figure 2: Candidate presentation/code sensitivity. Left: bars show repeat-to-reversal regressions (correct to incorrect) and corrections (incorrect to correct), whose difference is the net accuracy effect. Right: open markers show original-to-repeat flips and filled markers repeat-to-reversal flips with pointwise 95% paired cluster intervals. The control sets contain 100/100, 100/100, 36/18 and 95/95 decisions/source clusters respectively. Banking77 and MASSIVE use dataset references; JevBench and ReflexBench use authored or AI-reviewed references. No averaging across sources is performed.

Policy	Model	Action	Review	Severity	Majority	Joint
Access	Laya EN	34.4	44.8	43.8	34.4	2.1
Access	Laya Multi	29.2	40.6	29.2	34.4	0.0
Access	Llama 8B	34.4	40.6	68.8	34.4	0.0
Access	Qwen 8B	66.7	53.1	79.2	34.4	46.9
Access	Jev 1.13	100.0	89.6	95.8	34.4	100.0
Refund	Laya EN	25.0	77.1	31.2	25.0	0.0
Refund	Laya Multi	25.0	74.0	31.2	25.0	0.0
Refund	Llama 8B	33.3	74.0	59.4	25.0	0.0
Refund	Qwen 8B	25.0	74.0	43.8	25.0	1.0
Refund	Jev 1.13	100.0	100.0	100.0	25.0	100.0
Routing	Laya EN	71.9	69.8	35.4	25.0	28.1
Routing	Laya Multi	24.0	63.5	33.3	25.0	0.0
Routing	Llama 8B	25.0	63.5	59.4	25.0	0.0
Routing	Qwen 8B	49.0	64.6	79.2	25.0	28.1
Routing	Jev 1.13	54.2	100.0	95.8	25.0	29.2

Table 2: Fresh policy reference accuracy (percent). Joint requires both original and one-field-counterfactual actions to be correct. Majority is the original action-label baseline. All rows have 96 paired base clusters; the complete machine-readable analysis includes pointwise cluster intervals and all reference-label distributions.

such cells can reflect consistently wrong decisions. We report base accuracy and joint counterfactual success alongside every stability diagnostic to expose this ambiguity.

Representation effects also depend on the supplied policy. Jev’s authored Chinese routing rendering increases action accuracy by 45.8 percentage points, with a within-family simultaneous interval of [31.4, 60.2]; the underlying English facts and label IDs remain fixed. This is a difference between two complete policy/question/criteria renderings, not a general Chinese-language advantage. Laya English routing loses 46.9 points under explicitly irrelevant archived records. The full effect family, including null and opposite-direction results, appears in Figure 13.

A post-hoc routing inspection localizes all 44 Jev English action errors to states where exactly one of outage and paying is true: every such state is routed to urgent support, although the policy requires both. All 52 states with matching boolean values are correct. The Chinese rendering is correct on all four boolean combinations. This behavioral pattern motivated the held-out wording test below; it did not isolate the cause because the original language intervention changed multiple text fields together.

The LLM codebook control shows why ordinary candidate reversal can be misleading (Figure 3). For Llama access decisions, reversing display alone changes no predictions, changing only the label-to-code assignment changes 55.2%, and changing both again changes none. On routing, the corresponding values are 0%, 62.5% and 3.1%. These are observed non-additive effects within the fixed adapter: apparent stability under the combined intervention can coexist with sensitivity to code identity. They do not identify an architectural mechanism. The codebook run is a separate inference block, so its repeat and both-change outcomes are not assumed identical to the earlier batch context.

4.5 Held-out wording: a proposed explanation fails to generalize

The new routing cases balance all four outage/payment truth combinations by construction. Figure 4 reports the explicit-minus-compact wording effect on the 48 exclusive-or states, where erroneously interpreting “or” for “and” changes the action. Jev improves from 7/48 to 13/48 in English (+12.5 percentage points; paired 95% CI [2.1, 25.0]), but falls from 46/48 to 13/48 in Chinese (−68.8 points; CI [−81.3, −56.3]). The explicit conjunction is therefore not a language-independent repair. Qwen improves by 29.2 points in English and 52.1 in Chinese, while Laya English worsens by 31.2 and 12.5 points respectively. We report these mixed results because a universal “clearer wording helps” account would be contradicted by the held-out data. The experiment changes one authored policy clause within each language; it does not identify whether the cross-language difference arises from translation quality, tokenization or model training.

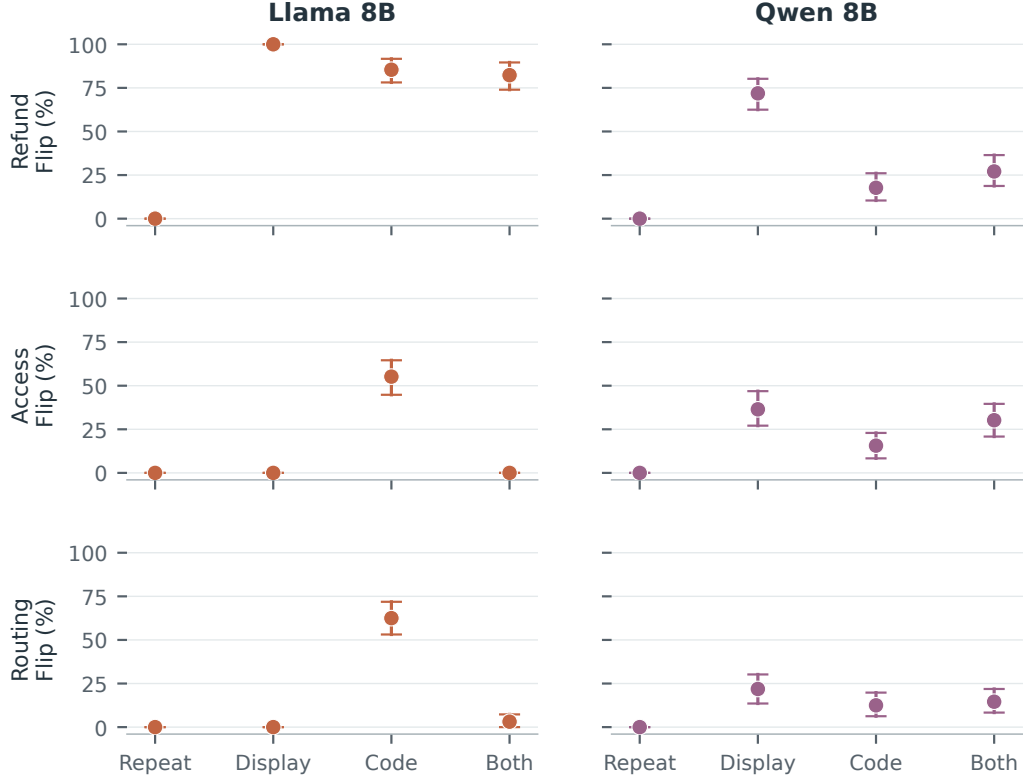


Figure 3: Fresh LLM codebook controls: 96 base action decisions per policy and model, with pointwise 95% paired cluster intervals. Display reverses semantic option order with code identity fixed; Code reverses code identity with semantic order fixed; Both changes both. Ground truth is an executable supplied policy. The code-only sensitivity can be hidden by the combined reversal.

5 Cross-domain error complementarity

Historical cross-domain scores can conceal two different properties: whether the systems make *complementary* errors, and whether their *agreement* is trustworthy. We examine all 28 main suites without pooling across their 15 source collections or reference standards. For a suite s , let c_{im} denote reference agreement of system m on typed decision i . We define the best observed single-system rate $b_s = \max_m \text{mean}_i c_{im}$ and the reference-aware oracle ceiling $u_s = \text{mean}_i \max_m c_{im}$. Their difference $u_s - b_s$ is *selection headroom*, not the gain of a trained or deployable router: the oracle uses each decision’s reference to choose a system. Separately, a strict five-system majority exists only when at least three valid outputs are the same label. We report its coverage and conditional error risk, retaining failures in the fixed five-system panel. Pointwise intervals resample source/state clusters within each suite.

Figure 5 reveals why the two properties should not be conflated. On SST-5 with the choice interface (dataset-provided reference), the best observed system reaches 55.9% agreement and the oracle ceiling 87.3%, a 31.4-point gap (95% cluster interval [28.5, 34.4]); nevertheless, strict majority covers only 78.6% of decisions and errs on 42.7% of those covered (interval [39.3, 46.1]). On Aegis2 prompt annotations (human-reference track), oracle headroom is 16.0 points [14.4, 17.7], while majority covers all decisions but has 26.0% error [24.1, 28.0]—8.4 points less reference agreement than the best observed system (paired interval [−10.5, −6.4]). The opposite geometry also occurs: on the programmatic Jev–Laya needle suite, oracle headroom is only 0.3 points [0.0, 0.7] while majority risk is 21.3% [13.0, 30.1] at 99.8% coverage; that suite has 900 decisions but only 25 original content clusters. A high oracle ceiling thus identifies *potential* error complementarity, not a consensus-based way to realize it. Because system selection, examples and analytical questions were developed after

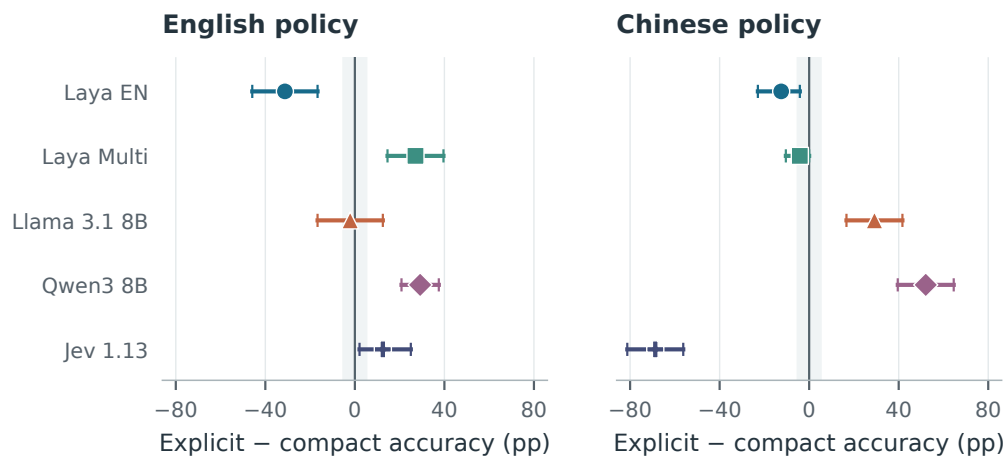
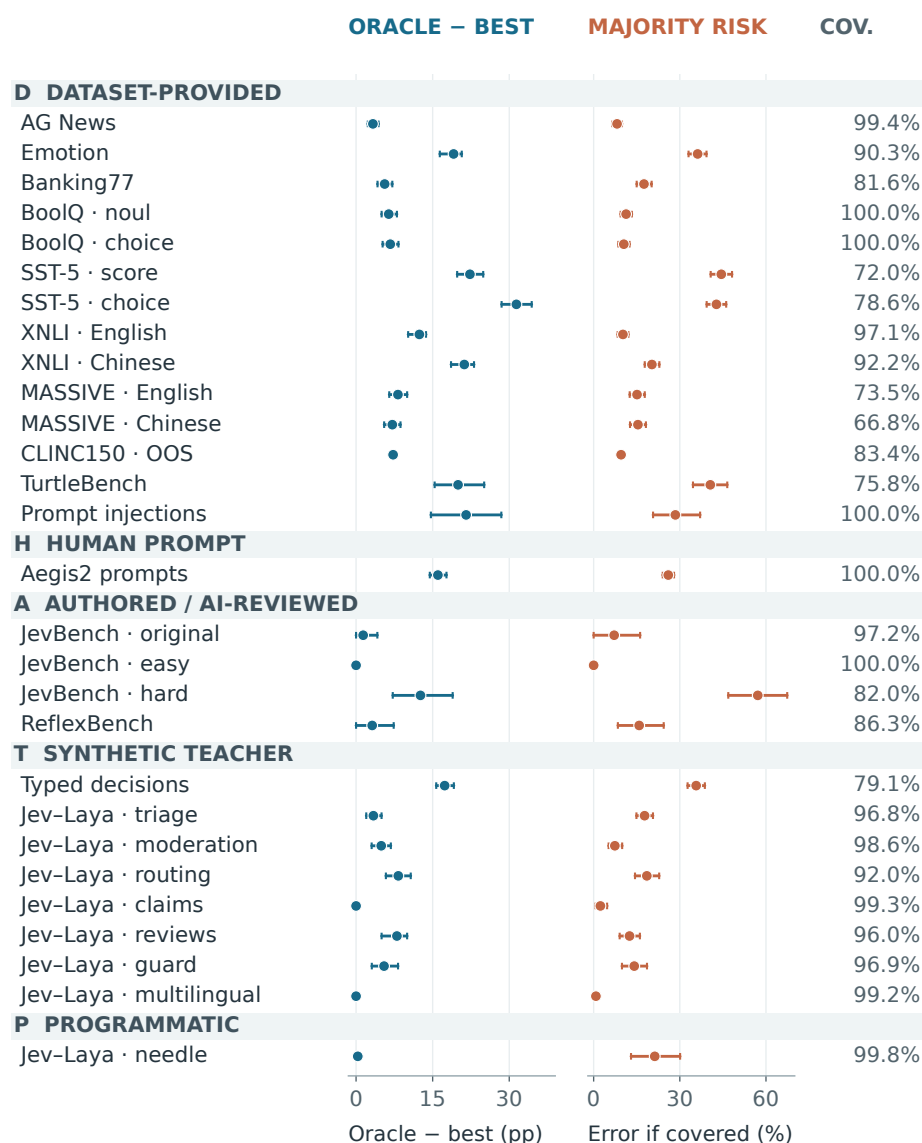


Figure 4: Held-out routing wording challenge on 96 new base states (48 XOR cases). Points are paired explicit-minus-compact action-accuracy effects; whiskers are pointwise 95% bootstrap intervals stratified by generator seed and truth cell. Within each language only the first policy clause changes. English is the predeclared primary contrast; Chinese is secondary. The shaded band marks a ± 5 -point descriptive region, not a significance threshold.

the historical outputs existed, these are exploratory within-suite descriptions, not evidence that a cross-domain selector would generalize.

Complementary errors, fragile consensus

28 suites · 5 systems · pointwise 95% state-cluster intervals



Oracle uses each reference: upper bound, not a deployable selector.

Majority: ≥ 3 matching valid labels; risk is conditional on coverage.

Figure 5: Selection headroom and strict-majority risk across 28 main suites, grouped by reference provenance: D, dataset-provided; H, human prompt; A, authored or AI-reviewed; T, synthetic teacher; P, programmatic. Left: the reference-aware oracle ceiling minus the best observed system, *not* an attainable router gain. Right: error conditional on at least three identical valid labels, with coverage in the adjacent column. Whiskers are pointwise 95% state-cluster bootstrap intervals (5,000 draws). Shared-source variants are not independent; no across-source inference is made.

6 A matched intervention in legal and scientific evidence

To test whether the historical suite atlas yields a source-independent stability conclusion, we separately froze a new, model-blind diagnostic from *independent* natural-domain references: 144 document-hypothesis decisions from 91 ContractNLI test contracts [Koreeda and Manning, 2021], and 120 claim-cited-abstract decisions from SciFact’s labeled development set [Wadden et al., 2020]. The legal sample is balanced across entailment, contradiction and not-mentioned; the scientific sample is balanced across annotated support and contradiction. Each source pair receives an unchanged request, a byte-identical repeat, and a reversal of the candidate dictionary’s insertion order. State, candidate meanings and reference label are invariant; no gold or source ID is sent to the models. All 792 requests were independently audited as complete for the four local adapters before inference; Jev’s HTTP payload is complete, but server-side tokenization is not observable. We resample by contract or claim for pointwise 95% intervals, and never combine the two domains into a prevalence-weighted score.

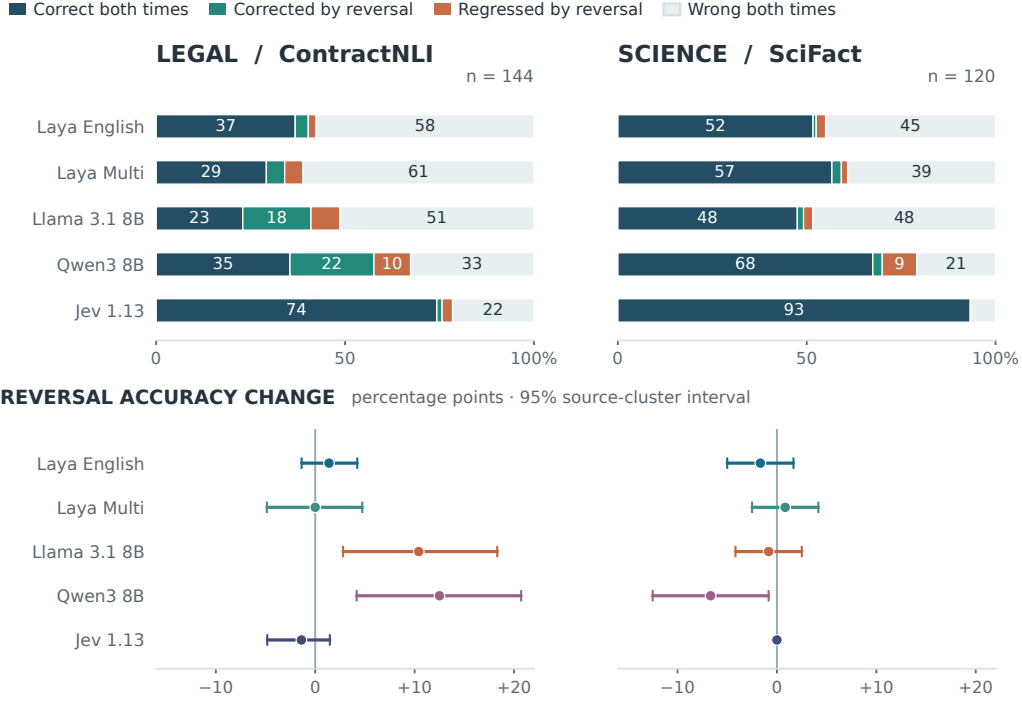
The paired state decomposition in Figure 6 exposes information an accuracy table would discard. On legal inference, the two code-likelihood LLM adapters change 66/144 (Llama) and 64/144 (Qwen) selected labels under ordinary reversal. Llama’s reference accuracy rises from 44/144 to 59/144, and Qwen’s from 65/144 to 83/144, yet only 33/144 and 51/144, respectively, are reference-correct under *both* renderings. These apparent gains are the net of 26 corrections versus 11 regressions for Llama and 32 versus 14 for Qwen. Their paired gains are +10.4 points (cluster interval [+2.8, +18.3]) and +12.5 points ([+4.2, +20.7]), respectively; they should not be read as better legal reasoning after a semantic change. In these adapters reversal also reassigns answer codes, so it is a joint model-interface perturbation; Section 8 isolates its factors.

The same intervention has a different geometry on scientific evidence. Qwen changes 14/120 labels and loses eight net correct answers (92/120 $\rightarrow$ 84/120; paired -6.7 points, $[-12.5, -0.8]$), while Jev remains reference-correct on the same 112/120 pairs under both orders. Jev’s legal original/reversal agreement is 138/144, versus 142/144 on a literal repeat; all four local deterministic adapters have zero literal-repeat flips on these requests. Thus even the direction of the *net* accuracy effect varies by domain and adapter, while joint correctness captures a more stable operational requirement. These are selected-pair descriptions, not a claim that the five systems’ legal or scientific abilities are ranked across their native interfaces.

Scope matters. The legal 16,000-character feasibility cap excludes 24/123 official test contracts and is format-selective; 48/144 contradiction cases in the selected diagnostic do not reflect the official test label prevalence. ContractNLI repeats 17 fixed hypotheses across documents. A document-blind lookup trained solely on the official training contracts’ modal label for each hypothesis already attains 82/144 (56.9%) on our frozen legal sample; it is a supervised shortcut diagnostic, not a fair zero-shot model baseline. SciFact’s positive-evidence subset excludes no-information decisions and does not test retrieval; some contradiction claims originate from deliberate inversion of source claims. Source annotations are treated as references, not independently re-adjudicated truth. We report format/length and label-prior audits in Appendix B, and retain source attribution and reconstruction instructions in the companion artifact.

PAIRED CORRECTNESS / INDEPENDENT NATURAL DOMAINS

Same source pair · exact repeat · candidate reversal · five fixed model-adaptor systems



Legal: 144 balanced document-hypothesis pairs (91 contracts). Science: 120 positive-evidence claim-abstract pairs. LLM reversal changes display order and answer-code mapping jointly; strict model/interface effect only. Exact-repeat flips: 0 for four local systems; Jev legal 2/144, Jev science 0/120. Error/coverage audited separately.

Figure 6: Paired correctness under candidate reversal on two independently sourced, deliberately class-balanced diagnostics. Each horizontal bar partitions the same source pairs into correct under both renderings, corrected, regressed and wrong under both; bars sum to 100%. Lower panels show reversal minus base reference accuracy with pointwise 95% contract- or claim-cluster intervals (10,000 draws). The LLM intervention changes display and code assignment jointly; codebook controls are separate. SciFact includes only positively annotated cited abstracts, not retrieval or no-information examples.

7 Cited but inconclusive scientific evidence

The positive-evidence SciFact slice in Section 6 cannot test whether a system distinguishes evidential silence from support or contradiction. We therefore froze a *separate* three-choice claim-cited-abstract diagnostic from SciFact’s labeled development files [Wadden et al., 2020]. The official schema defines `cited_doc_ids` as papers named in the claim’s source citation and records abstract evidence only where found.¹ Across 339 unique cited pairs, there are 138 SUPPORT, 71 CONTRADICT and 130 cited papers without an evidence entry; only the last group is eligible for our derived NOINFO reference. Arbitrary unannotated corpus pairs are never relabeled as negatives. A model-blind hash selected 60 pairs per class, with 180 distinct claims and 180 distinct papers. All received an original request, byte-identical repeat, reversed criterion order and a predeclared removal of only the paper title (720 requests per system). All local adapter requests passed complete-token audits; the hosted HTTP requests were complete, although server-side tokenization is not observable.

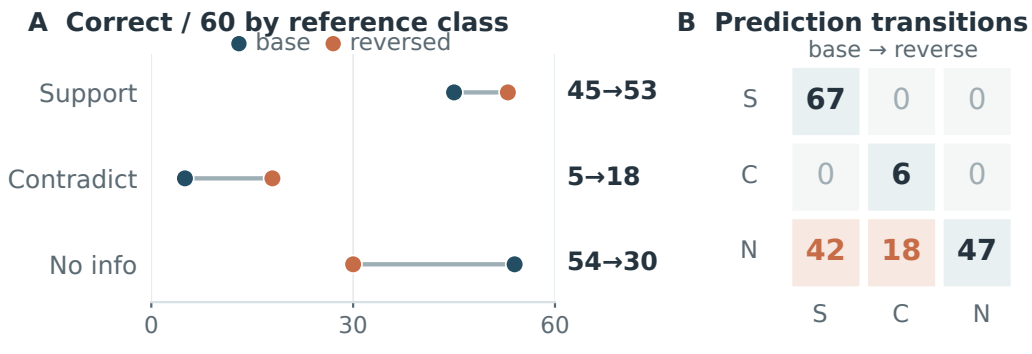
Figure 7 shows why the net score is insufficient. Under option reversal, Qwen changes 60/180 semantic answers, versus 0/180 on exact repeat, but its total reference-correct count barely moves (104 → 101/180; paired −1.7 percentage points, 95% cluster-bootstrap interval [−7.8, +4.4]).

¹<https://github.com/allenai/scifact/blob/master/doc/data.md>

System	Base correct	Both correct	Reverse flips	No-title flips
Laya English	81	71	16	60
Laya Multilingual	73	59	38	68
Llama 3.1 8B	78	61	42	18
Qwen3 8B	104	80	60	16
Jev 1.13	148	148	0	1

Table 3: Three-class SciFact diagnostic on the same 180 source pairs (60 per reference label). “Both correct” requires correctness under both original and reversed criteria; flips count changed semantic predictions among valid pairs. All responses were valid. Exact-repeat flips were zero for the four local systems and 2/180 for Jev. These are model–adapter outcomes, not an end-to-end retrieval ranking.

QWEN3 8B / 180 CITED CLAIM-ABSTRACT PAIRS Near-flat accuracy, 60 changed semantic answers



Total correct 104→101/180; Δ -1.7 pp (paired 95% CI -7.8 to +4.4)

60/180 reversal flips vs 0/180 exact-repeat flips. S/C/N: support/contradict/no-info.

Figure 7: Qwen’s hidden cancellation on the balanced three-class SciFact diagnostic. Left: reference-correct counts by class on the same 60 pairs; right: complete 3×3 semantic prediction transitions across all 180 pairs (S/C/N denote SUPPORT/CONTRADICT/NOINFO). Overall correctness changes little despite 60 label switches, all away from baseline NOINFO predictions. The paired interval uses 10,000 label-stratified claim/document-cluster resamples; exact-repeat switches were zero. NOINFO is restricted to an explicitly cited abstract without annotated evidence.

Class-specific effects oppose one another: SUPPORT rises 45 → 53/60, CONTRADICT 5 → 18/60 (+21.7 points, [+11.7, +31.7]), while NOINFO falls 54 → 30/60 (−40.0 points, [−53.3, −28.3]). All 60 changed predictions had been NOINFO at baseline: 42 move to SUPPORT and 18 to CONTRADICT. Thus the near-flat total conceals an answer-marginal migration, not stable decisions. For this constrained code-likelihood adapter, reversal also changes the code–label assignment; the observation is a joint *interface* effect, not a pure semantic-order effect (cf. Section 8).

Title removal is a distinct sensitivity probe. For Laya English, it changes 60/180 labels and increases NOINFO reference-correct decisions from 20 → 39/60 (+31.7 points, [+18.3, +45.0]); the overall gain is +10.6 points ([+3.9, +17.2]). Other systems do not share a consistent gain (Table 3). Removing a title also changes state serialization, while the frozen question still mentions that a title identifies the paper even when the field is absent; this is a visibility stressor, not proof of a lexical shortcut or a causal estimate of abstract reliance. The source’s 138/71/130 candidate distribution was deliberately rebalanced, its annotations were not independently re-adjudicated, and no experiment here tests retrieval or full papers. The earlier two-choice positive-only SciFact slice overlaps this sample on only 28 SUPPORT and 55 CONTRADICT pairs; changed questions, label space and selection preclude a direct accuracy comparison. Source and input audits are detailed in Appendix B and the companion artifact.

7.1 Sensitivity to the balanced SciFact selection

The preceding 60/60/60 slice deliberately changed the official cited-pair label mix. To test whether its class-conditional reversal signal depends on that selection, we conducted a *post-hoc same-source census*: every one of the 339 eligible development claim–cited-abstract pairs (138 SUPPORT, 71 CONTRADICT, 130 derived NOINFO) received all four three-choice conditions in a fresh run of all five systems. The 159 pairs outside the prior three-choice selection contain 78/11/70 by reference class; they are not an independent holdout, and 37 positive pairs had already appeared in our earlier *two-choice* SciFact task. Claims and documents can recur in the census, so paired 10,000-draw percentile intervals resample claims as the primary cluster and documents as a sensitivity check. The NOINFO reference retains its narrow provenance: an explicitly cited document with no annotated abstract-evidence entry, not an independently re-adjudicated negative.

System	Full 339	Outside 159	NOINFO 70	Flips / repeat
Laya English	171→164	90→88	16→13	29 / 0
Laya Multilingual	150→150	77→80	8→8	63 / 0
Llama 3.1 8B	182→158	103→90	33→15	73 / 0
Qwen3 8B	231→220	129→119	60→38	97 / 0
Jev 1.13	291→292	144→143	59→59	5 / 4

Table 4: Fresh full-census results. Each arrow shows base→joint-reversal reference-correct counts on the stated denominator. “Outside 159” means outside the earlier balanced *three-choice* selection, not unseen model data. Flips compare valid semantic labels; repeat is byte-identical to base. All base/reversal outputs were valid; Jev had one invalid title-removal response, retained separately.

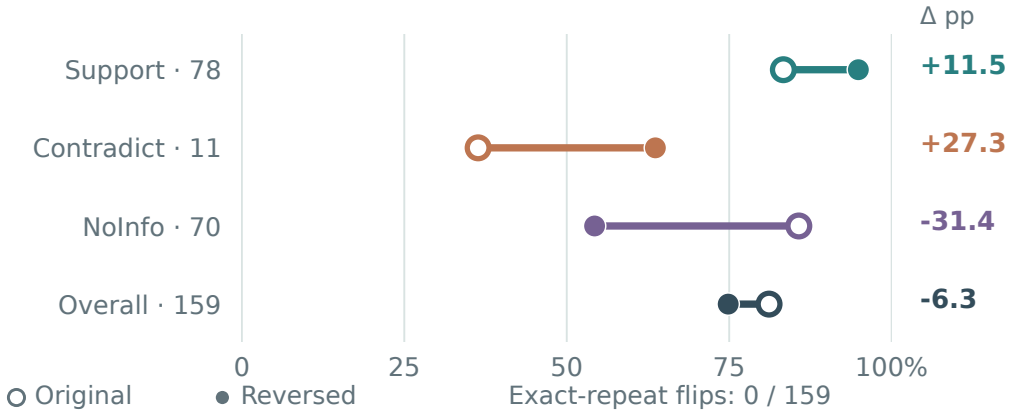
NoInfo loss survives the balanced-sample selection

SciFact claim–cited-abstract pairs · Qwen3 8B · joint option reversal

REFERENCE MIX



QWEN / 159-PAIR THREE-CHOICE COMPLEMENT



NoInfo = cited abstract without annotated evidence; not adjudicated neutrality.

Figure 8: Fresh, post-hoc SciFact cited-pair census sensitivity check. The top strip compares reference-label composition in the selected 180, its 159-pair complement, and all 339 eligible pairs. The lower panel shows Qwen’s original (open) and joint-reversal (filled) reference agreement on the complement; differences are percentage points. This is the same source, not an independent holdout: 37 complement positives appeared in the earlier two-choice task. NOINFO is an explicitly cited abstract without annotated evidence, not adjudicated neutrality.

The earlier Qwen class-opposing pattern persists in the previously unselected three-choice pairs: on their NOINFO references, correctness drops $60 \rightarrow 38/70$ (-31.4 points; paired claim-cluster 95% interval $[-42.4, -21.1]$, document-cluster $[-42.9, -20.3]$), with 22 correct NOINFO decisions migrating to other labels and no reverse corrections into NOINFO. SUPPORT improves $65 \rightarrow 74/78$ and the much smaller CONTRADICT slice $4 \rightarrow 7/11$, leaving the aggregate $129 \rightarrow 119/159$ (-6.3 points, claim interval $[-13.7, +1.2]$). On the full 339 pairs, Qwen’s NOINFO is $112 \rightarrow 68/130$ (-33.8 points, claim interval $[-42.1, -26.0]$), while overall correctness is $231 \rightarrow 220/339$ (-3.2 points, $[-8.4, +2.0]$). Its full-census prediction marginal changes from SUPPORT/CONTRADICT/NOINFO $144/11/184$ to $216/33/90$; thus the near-flat aggregate again conceals a substantial answer redistribution. Llama also loses NOINFO correctness on the 159-pair complement ($33 \rightarrow 15/70$, claim interval $[-36.8, -15.7]$), whereas Laya and Jev do not show the same class-specific magnitude (Table 4). These are joint interface effects, not isolated code or display-order effects.

An identical-request cross-run audit tempers exact-count comparisons with the balanced run. All 720 overlapping request hashes match; for the two local LLMs, every overlapping prompt-token hash matches as well. Nonetheless, Qwen’s prior balanced base has two NOINFO decisions that change to SUPPORT in the fresh census run, shifting its selected-subset base count $104 \rightarrow 102/180$; reversal remains $101/180$. The fresh selected-subset NOINFO decline is still $52 \rightarrow 30/60$ (claim interval $[-48.3, -25.0]$), versus $54 \rightarrow 30/60$ previously. Both used batch size four but different co-batches; this audit cannot isolate batching from other numerical or serving variability. We therefore never splice prior selected-subset predictions into the new complement. One Jev InvalidDecision occurred only under title removal; all other census responses were valid. A separate, offline semantic tie-break sensitivity changes full-census reversed correctness only from $158 \rightarrow 159/339$ for Llama and $220 \rightarrow 219/339$ for Qwen; the class-level direction persists, but these ties reinforce that the decoder is part of the evaluated adapter. The census shares the same source and derived-reference limitations as Section 7; the title-removal question’s reference to an absent title also precludes a clean causal interpretation. Complete confusion matrices, claim/document intervals, case-level predictions, overlap IDs and SHA-256-bound raw outputs are in the companion census artifact.

7.2 Execution-context stability is not interface invariance

Motivated by the cross-run drift above, we froze a new follow-up before its own inference: all 339 census pairs in both base and joint-reversal conditions, two local LLMs, and five execution plans (6,780 decisions). The plans are original contiguous batches of four, an identical repeat in the same process, reversal of row order within the same batches, seeded reassignment to new batches of four, and single-request execution. No case is selected by its reference, prior error or logit margin. The unchanged adapter uses the pinned weights, chat template, BF16/SDPA and explicit position IDs; every unpadded prompt-token and request hash matches across plans and against the prior census. Only execution context changes, not the model-facing candidate interface.

Both models have zero semantic flips under identical-batch repeat and within-batch row reversal. Repacking changes 1 base and 4 reversed-condition Qwen predictions, versus 3 and 5 for Llama. Singleton execution changes 3 and 3 Qwen predictions, versus 7 and 5 for Llama, respectively; each count has denominator 339 (Figure 9). Original top-two candidate-logit margins of these changed decisions are at most 0.5 for Qwen and 0.375 for Llama; several are exact BF16 ties. Native tie decoding is retained. The original-plan semantic labels reproduce the previous full census for both models. Nonetheless, across all five plans, joint option reversal still changes 95–97 Qwen and 71–73 Llama predictions. The derived NOINFO reference-agreement difference stays between -34.6 and -33.1 percentage points for Qwen and between -27.7 and -26.9 for Llama. Each plan’s paired claim-cluster 95% interval excludes zero; these are pointwise descriptive intervals on the fixed source, not independent replications across plans.

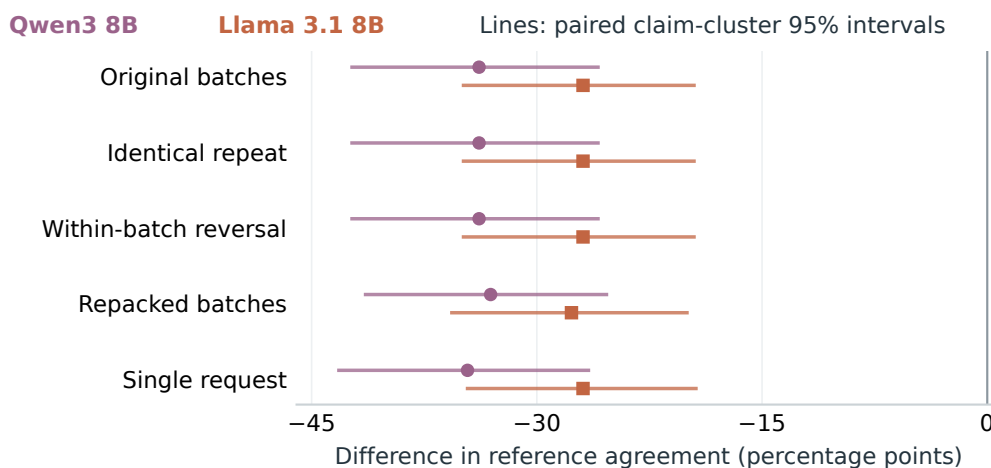
EXECUTION CONTEXT / SCIENTIFIC EVIDENCE

339 identical inputs per condition; 5 plans; 2 fixed local adapters

a Prediction flips vs. original batches (count / 339)

	Qwen3 8B		Llama 3.1 8B	
	Base	Reversal	Base	Reversal
Same batches, repeat	0	0	0	0
Same members, row reversal	0	0	0	0
New co-batches, size 4	1	4	3	5
Single-request execution	3	3	7	5

b NOINFO reference agreement: reversal minus base (pp)



NOINFO: 130 derived cited-annotation-absence references; not adjudicated neutrality.

Fixed plan order; same-source follow-up; no isolated execution-mechanism claim.

Figure 9: Execution-context follow-up on all 339 SciFact cited pairs. (a) Same-condition semantic flips relative to original contiguous batches; columns distinguish base and joint option reversal. (b) Reversal-minus-base reference agreement on all 130 derived NOINFO pairs, with 10,000-draw paired claim-cluster percentile intervals. Execution perturbations move few predictions while the class-specific interface effect persists. NOINFO is cited annotation absence, not adjudicated neutrality. Fixed execution-plan order is not counterbalanced.

This control supports robustness of the class-level interface pattern to the tested execution contexts, not bitwise determinism or a unique numerical mechanism. Repacking and singleton execution also change padding and kernel shapes; the fixed plan order does not isolate time-dependent effects. Same-process repeats do not establish cross-process, cross-GPU or cross-version reproducibility. These new outputs remain separate from historical results in `research/batch_context_v1/`.

Reference audit prepared, not completed. A source-verified blinded work package includes all 130 derived NOINFO pairs plus 15 annotated support and 15 contradiction calibration pairs. Two independently randomized packets omit reference labels, strata and model outputs; independent annotation forms are blank. The workflow retains four-way agreement (including UNCERTAIN), separate derived/calibration statistics, locked form hashes and third-reviewer adjudication. Source-containing packets remain restricted and precisely ignored by version control. No human-validity result is claimed until independent reviewers complete this work.

8 Separating candidate order from answer-code assignment

The legal reversal in Section 6 is a useful stress test, but it is not a single-factor manipulation for the two code-likelihood LLM adapters: reversing the serialized candidates also changes which semantic label is assigned to each answer token. We therefore completed a factorial follow-up on the *same* 144 ContractNLI cases from 91 source contracts. The original and joint-reversal outputs are reused by hash; two missing cells reverse only display order or only label-to-code assignment. The state, option meanings, gold label, model weights and adapter are fixed. Before inference, the four rendered messages and code maps were frozen and the reused cells were checked against their historical tokenized-prompt hashes. A transparent corrective amendment aligned the new inference batch size to the historical four after an initial batch-eight pilot had already been inspected; both pilots and all raw outputs remain available. This is a follow-up diagnostic of these two pinned model–adapter pairs, not a wholly prospective four-cell test or an intervention on legal reasoning itself.

Order and answer-code mapping have different effects

144 matched ContractNLI pairs/model · fixed semantic tie-break · batch 4

Llama 3.1 8B

Qwen3 8B

	BASE MAP	SWAP MAP		BASE MAP	SWAP MAP
Orig.	30.6% BASE	34.7% 125 flips	Orig.	45.1% BASE	43.8% 10 flips
Rev.	33.3% 6 flips	39.6% 54 flips	Rev.	54.2% 36 flips	58.3% 69 flips

Cells show accuracy; flips count changed semantic labels vs each model's base.

Figure 10: Orthogonal ContractNLI candidate-rendering diagnostic for Llama 3.1 8B and Qwen3 8B ($n = 144$ matched document–hypothesis cases/model). Columns change semantic label-to-code assignment; rows change serialized display order. Cell accuracy and prediction switches from base use a *common semantic tie-break* for the 2×2 comparison; this is a labeled post-hoc sensitivity, not a replacement for the historical native outputs. All cells are evaluated at historical batch size four; intervals in the text resample the 91 contracts. The source sample is deliberately class-balanced.

Figure 10 shows the common-tie comparison. The native recorded accuracy grid (base, display-only, code-only, both) is 44, 48, 50, 59 of 144 for Llama and 65, 78, 63, 83 for Qwen; all 1,152 cell decisions are valid. Yet the single-factor *prediction* transitions have nearly opposite geometry. Changing only the answer-code mapping flips 125 Llama labels (86.8%; contract-cluster 95% interval [81.1, 92.1]) but only 10 Qwen labels (6.9%; [2.8, 11.6]); changing only display order flips 6 Llama labels (4.2%; [1.4, 7.6]) but 36 Qwen labels (25.0%; [18.1, 32.2]). The selected code itself reveals why these are not equivalent perturbations: Llama selects token A in 138/144 base and 131/144 code-only cases despite the remapping, whereas Qwen shifts from token C in 122/144 base cases to token A in 126/144 code-only cases while mostly retaining its *semantic* answers. This is evidence of distinct interface coupling in the two implementations, not a general property of either model family.

There is a third interface degree of freedom: the historical joint-reversal question also reverses the decoder's first-maximum tie priority. Exact BF16 top-logit ties occur in Llama's base/display/code/both cells 4/0/9/12 times and Qwen's 0/4/0/5 times. Re-decoding stored probabilities with one fixed semantic priority (entailment, contradiction, not-mentioned) changes only the historical joint cell: Llama 59 $\rightarrow$ 57 correct and 66 $\rightarrow$ 54 base-to-joint flips; Qwen 83 $\rightarrow$ 84 correct and 64 $\rightarrow$ 69 flips. We retain the native counts above for reproducibility, but interpret the factorial grid only under this common, explicitly post-hoc tie rule. Under that convention, the joint accuracy effects are +9.0 percentage points for Llama (paired contract-cluster interval [+2.1, +16.1]) and +13.2 for Qwen ([+4.4, +21.4]). Their difference-in-differences interactions are +2.1 ([−7.0, +10.8]) and

+5.6 points ($[-2.1, +13.0]$), respectively; these intervals do not resolve an interaction. The original exact-repeat output has zero prediction changes in either model. Conversely, the batch-eight pilot changes three Qwen display-only predictions versus the matched-batch result, reinforcing why batch shape cannot be quietly varied. All intervals describe this selected, balanced, length-capped legal sample, not ContractNLI’s natural test distribution or system-level legal competence.

9 A non-additive codebook effect on scientific evidence

The three-class SciFact diagnostic supplies a deliberately different domain for the orthogonal control in Section 8. We use the same 180 frozen claim–cited-abstract pairs, with 60 cases per operational label and one selected pair per source claim and document. The already observed base and ordinary joint-reversal outputs are reused by hash; display-only and code-only cells were frozen before their own inference. All four prompts were audited, including exact tokenizer-token equality of the reused cells with their historical records. New inference matches the original batch size four, checkpoint, chat template and code-likelihood adapter. This is an exploratory missing-cell follow-up, not a wholly prospective discovery or a test of retrieval.

The joint reversal is not the sum of its parts

180 matched SciFact3 cited-abstract pairs/model · common semantic tie-break

Llama 3.1 8B

	BASE MAP	SWAP MAP
Orig.	43.3% BASE	39.4% 85 flips
Rev.	33.9% 51 flips	37.8% 46 flips

Qwen3 8B

	BASE MAP	SWAP MAP
Orig.	57.8% BASE	63.3% 18 flips
Rev.	63.3% 30 flips	56.1% 60 flips

QWEN / DESTINATION OF THE SAME 107 BASE-NOINFO PREDICTIONS

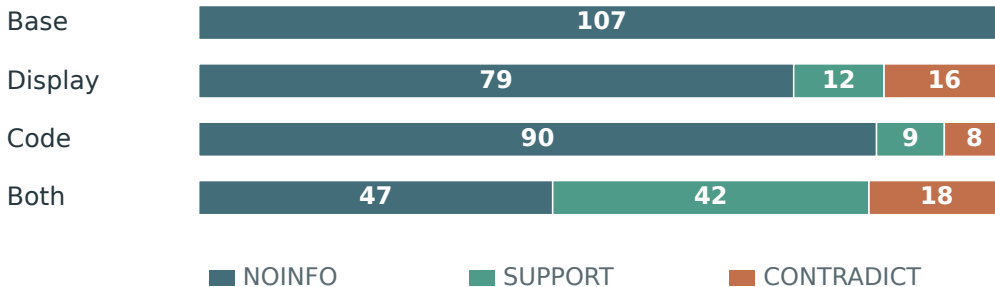


Figure 11: SciFact3 orthogonal display-order $\times$ label-to-code diagnostic on 180 matched cited-abstract pairs per model. The grid gives accuracy and semantic prediction flips from each model’s base under a common semantic tie-break; old native outputs remain unchanged. The lower bars follow the *same 107 cases* that Qwen predicts NOINFO at base, partitioned by their destination class in the other cells. Both factors together have a non-additive effect on this pinned model–adapter system; the figure does not identify a neural mechanism. NOINFO means no annotated abstract evidence for a cited paper, not no evidence anywhere.

Figure 11 makes the masking of class-level change visible. Qwen is correct on 104/180 base pairs, 114/180 in either single-factor cell, and 101/180 under the ordinary joint reversal. Display-only and code-only change 30 and 18 base predictions, respectively, but the joint intervention changes

60. Both single factors gain +5.6 accuracy points; their combination loses −1.7 points, for an accuracy difference-in-differences of −12.8 points (pointwise 95% paired source-claim-cluster interval [−19.4, −6.1]; 10,000 draws). The 60 joint flips *all* depart from the 107 cases Qwen labels NOINFO at base: display-only retains 79 as NOINFO and sends 12 to SUPPORT/16 to CONTRADICT; code-only retains 90 and sends 9/8; both retains only 47 and sends 42/18. Among 60 reference-NOINFO pairs, Qwen’s correct counts are 54/48/52/30 across base/display/code/both; among 60 reference-CONTRADICT pairs they are 5/18/12/18. Thus some migration repairs contradiction errors while a larger joint migration also creates positive predictions on NOINFO-reference pairs. Neither display nor code assignment alone explains the full ordinary reversal effect. The corresponding legal single-factor flip counts are 36/144 and 10/144 for Qwen; this cross-domain contrast is descriptive, because source selection, label semantics and task wording differ.

Llama’s native accuracy grid is 78/61/71/69 of 180, with 51/85/42 base-relative label flips. Historical joint reversal also changes the decoder’s first-maximum tie priority. Exact top-logit ties occur in Llama’s base/display/code/both cells 6/1/11/6 times and Qwen’s 1/6/1/0 times. Re-decoding the stored semantic probabilities with a fixed SUPPORT-CONTRADICT-NOINFO priority changes only six Llama joint predictions: its common-tie grid is 78/61/71/68, with 51/85/46 flips. Qwen’s native and common-tie four-cell results are identical, so its non-additivity is not a tie-breaking artifact. The historical exact-repeat controls have zero flips; all 720 new cell decisions are valid. We retain the native records as primary historical outputs and interpret the 2×2 prompt/code contrast under the fixed decoder rule. The selected pairs are class-balanced and the NOINFO reference is derived from absent annotated *abstract* evidence for a cited document; the pointwise intervals do not describe SciFact’s natural prevalence or end-to-end fact-checking.

10 Generative adapters expose model–interface interaction

The historical Llama and Qwen comparisons use a fixed one-forward-pass candidate-code logit adapter, not free generation. To test whether that choice drives the conclusions, we froze a prediction-independent, source-balanced challenge of 212 states from 18 suites spanning all 15 current source collections. One question per state, its reference, candidate order and full input are held fixed. CLINC150/OOS is deliberately balanced between six rejection and six named-intent cases; this is a diagnostic subset, not a prevalence-weighted benchmark estimate. Each model is evaluated with two additional zero-shot adapters: greedy direct code generation using the historical prompt, and greedy generation after a fixed brief-explanation instruction ending in a final code. The checkpoint and chat-template hashes match the original baseline. Neither prompted adapter emits calibrated typed confidence or an ordinal expectation, so this comparison concerns discrete label accuracy and decision turnover only.

We also froze a smaller, exact-input direct-generation control on the new natural-domain cases: the same 144 legal and 180 three-class scientific requests for each 8B model, with the historical message template, greedy decoding and a strict whole-response one-code parser. All 648 generated answers are valid and have exactly the same semantic label as the corresponding one-forward-pass code-logit outputs; no source-domain accuracy or prediction transition changes under this adapter substitution. Prompt-token hashes and source IDs match the original requests. This rules out a “single-token scoring versus direct code generation” discrepancy on these selected cases and this prompt; it does not show that either model’s best prompted legal/scientific performance has been reached, or that their probability outputs are equivalent.

Figure 12 separates two effects that an aggregate score would hide. Direct generation changes none of Qwen’s 212 predictions and only five of Llama’s, consistent with the historical adapter’s constrained next-token decision on this subset. Brief deliberation changes 55 Llama and 46 Qwen predictions. Llama’s correctness rises from 133/212 to 145/212: 25 incorrect decisions are corrected while 13 correct decisions regress, with three invalid generations. Qwen moves from 158/212 to 153/212: 12 corrections coexist with 17 regressions and ten invalid generations. The selected-state paired effects are +5.7 points (95% interval [+0.5, +10.8]) and −2.4 points ([−7.1, +2.4]), respectively. However, when the 15 sources receive equal weight, Llama’s effect is +4.7 points with interval [−1.4, +10.6]; a source-general improvement is not established.

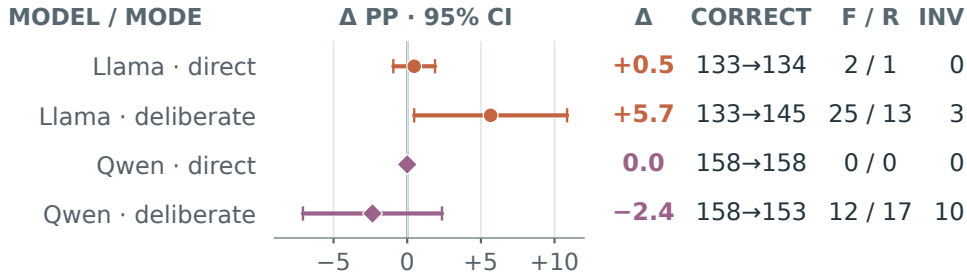
The source view makes the interaction more concrete without elevating small cells into claims of significance. Llama corrects four additional English XNLI cases but loses one Chinese XNLI

SYSTEM1BENCH / GENERATIVE CONTROL

Prompting changes who is right

Strict one-code parsing · matched cases · no prompt search

A PAIRED OUTCOMES



B SOURCE HETEROGENEITY

DELIBERATION – CODE-LOGIT · ACCURACY POINTS

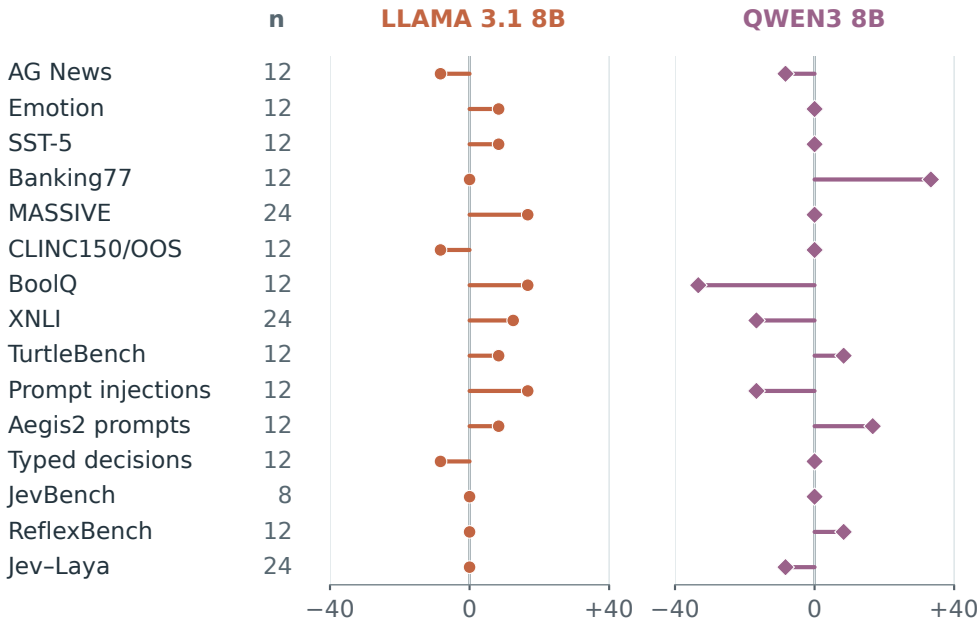


Figure 12: Prompted-baseline stress test on 212 matched states/model. Top: correction and regression flows, net accuracy effect and 95% paired suite-stratified bootstrap interval versus the historical code-logit output. Bottom: descriptive brief-deliberation effect by source ($n = 8-24$), on a shared percentage-point scale, without per-source significance tests. Invalid generations count incorrect; the post-hoc parser sensitivity is excluded.

case. Qwen corrects four Banking77 cases, whereas four BoolQ answers become invalid under the predeclared output grammar; its XNLI sample loses four correct answers in 24 states. These results

suggest that prompting is an intervention on a *model–adapter–task* combination, not an intrinsic improvement operator for an LLM.

Output validity materially affects the conclusion. Llama’s three invalids hit the 128-token cap before a final code. Eight of Qwen’s ten invalids actually end in an inline `FINAL: code` rather than the required standalone final line; two say `FINAL: None`. A clearly post-hoc end-of-response parser would recover eight answers, six correct, yielding 159/212 for Qwen instead of the registered 153/212. We retain the strict result as primary: the sensitivity distinguishes semantic selection from interface compliance, not permission to tune the evaluator after seeing answers. Brief deliberation used 87.6/90.1 seconds of synchronized local generation for Llama/Qwen versus 15.7/16.9 seconds for direct generation (batch size four, excluding load and tokenization); these implementation times are not merged with the separate performance experiment. One frozen 128-token prompt cannot bound either model’s best prompted performance.

11 Operational measurements and confidence ranking

11.1 Local quality and completion time

The local performance experiment evaluates 11 strata, four checkpoints, three batch sizes and eight rounds on the same A100 80GB PCIe, yielding 1,056 cells and 98,304 measured decisions. Each stratum uses 64 fixed measured states and disjoint warm-up states. Model processes are fresh per block; model order is balanced, CPU affinity and thread counts are fixed, and an external observer records GPU, worker-core and SMT-sibling activity.

The timing boundary includes uncached serialization, tokenization, transfer, inference and native structured-answer construction, with GPU synchronization. It excludes model loading, disk output, validation, network transport and queueing. Batch-one p50/p95 characterize request completion; batch-eight and batch-32 work divided by prediction time characterize fixed-batch throughput. Round statistics are summarized by their median and an eight-round bootstrap interval. Repeated states increase timing replication, not the effective accuracy sample size.

Figure 16 compares quality and latency within six declared workloads. For BoolQ, median-of-round request p50 values are 10.30, 8.55, 36.86 and 33.51 milliseconds for Laya English, Laya Multilingual, Llama and Qwen respectively. The quality coordinate belongs to the same fixed workload. It is not appropriate to infer a cross-task winner from the horizontal position alone: systems can trade reference agreement for lower completion time, and that trade-off changes with the task.

The host is shared. No foreign GPU compute process or sustained CPU/SMT threshold violation was observed, but six isolated residual-CPU intervals exceeded 15% without meeting the fixed two-interval rejection rule. These intervals remain included. All raw timing records, monitor samples and code/environment hashes are retained, and all 1,056 cells independently reconcile with the generated tables.

11.2 Selective prediction within a source

Confidence-based deferral and coverage–quality analysis are established evaluation tools, including for decision models [Geifman and El-Yaniv, 2017, Ibrahim and Zaki, 2026, Hu et al., 2026]. Figure 17 shows empirical accepted-set error as coverage grows under maximum class probability in our fixed local interfaces. All tied probabilities enter together, avoiding artificial improvements from an arbitrary within-tie ordering. The curve is computed separately for each model and source; intervals at displayed landmarks condition on the selected threshold.

These curves distinguish two failure modes that accuracy alone conflates: frequent errors at all confidence levels and errors concentrated among uncertain cases. They support a workload-specific investigation of escalation policies. A threshold selected from these test curves would be a post-hoc decision rule, so a deployable policy requires a separate calibration/development split and a held-out check. Candidate-code probabilities are evaluated as observed ranking statistics, not assumed calibrated equivalents of native decision confidence.

12 Limitations and scope

Reference and construct validity. The collection combines heterogeneous source references. Synthetic-teacher agreement, authored policy conformity and programmatic rule correctness are distinct targets. Static typed decisions do not measure interactive task completion or downstream utility. The fresh policies validate supplied-rule composition; their naturalness and Chinese realization have not received independent human adjudication. AG News and BoolQ belong to known Laya training-task families; these rows cannot establish unseen-task generalization. Other public-source training overlap is unknown.

Intervention and system scope. The local systems are fixed checkpoints with one declared adapter each. Output encodings can jointly change wording and computation, and candidate reversal changes presentation and LLM code assignments together. Findings describe these implementations rather than all decision-model or language-model architectures. Historical diagnostic selection is exploratory; the first generated cases supplied a pre-inference-frozen test of declared transformations. The held-out wording test was frozen after inspecting the first generated results but before its own inference; it tests a motivated hypothesis rather than independently discovering it. Its Chinese clauses lack independent human equivalence validation.

Operational scope. Local measurements use one shared host and fixed-batch inference, not optimized serving stacks or production arrival processes. Approximately one-hertz telemetry cannot exclude brief interference or shared memory-bandwidth effects. Hosted API version IDs and usage are observable, while server weights, tokenization, scheduling and resource allocation are not. The two timing boundaries remain separate.

Statistical scope. Matched variants are dependent and resampled together. Needle analysis has only 25 source clusters. Eight timing rounds support descriptions of these workloads, not rare-tail service guarantees. Pointwise exploratory intervals across many contrasts do not provide simultaneous paper-wide error control. Inference counts are execution totals, not independent-sample counts.

13 Conclusion

System1Bench measures reference agreement, joint correctness, correctness turnover, prediction stability and operational efficiency at a fixed typed decision interface. The source atlas and independent legal/scientific diagnostics keep workload type and reference provenance visible instead of turning unlike labels into a pooled score. On SciFact’s balanced three-way cited-pair diagnostic, Qwen’s near-unchanged total accuracy under candidate reversal conceals a large loss on no-information cases offset by gains on support and contradiction cases. A fresh post-hoc census of all eligible cited pairs shows the same class-level attrition outside that selected sample, while the aggregate difference remains imprecise; the derived no-information reference is not an adjudicated neutral class. The legal four-cell control goes further: superficially similar reversal outcomes arise from different couplings to displayed candidate order and answer-code identity in the two 8B adapters, with exact-score ties requiring their own sensitivity check. The supervised, document-blind ContractNLI prior and the mixed held-out wording test illustrate why source shortcuts and attractive post-hoc mechanisms must also be tested. Together these results suggest a practical reporting rule: preserve source-specific references, paired class transitions, orthogonal interface controls and the execution boundary. They characterize downstream decision behavior, not whole-system completion or deployment utility; those claims require evaluation of the full workflow.

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A Complete source-level results

Source	Laya EN	Laya Multi	Llama	Qwen	Jev
ag_news	94.3	92.4	88.9	86.8	88.6
emotion	59.1	53.0	50.5	54.8	59.0
banking77	55.3	51.2	54.1	66.8	79.9
massive_en	54.1	42.3	57.2	66.3	77.5
massive_zh	30.4	33.2	53.5	62.6	76.0
boolq	84.6	77.7	64.8	83.5	92.6
boolq_choice	83.6	77.4	68.7	83.3	92.2
xnli_en	86.0	81.7	46.2	78.7	85.9
xnli_zh	61.5	74.3	40.5	69.1	73.6
turtlebench	42.6	41.6	42.2	44.2	74.0
sst5	34.6	29.5	32.0	45.7	56.5
sst5_choice	49.6	35.6	42.2	43.9	55.9
prompt_injections	70.7	57.8	62.9	63.8	76.7
aegis2_prompt	49.6	57.1	66.8	72.7	82.4
typed_decisions	36.4	34.9	51.2	55.5	73.9
JB: original	70.8	41.7	75.0	83.3	98.6
JB: easy	95.8	89.6	100.0	100.0	100.0
JB: hard	29.7	32.4	34.2	46.8	72.1
ReflexBench	58.9	46.3	60.0	84.2	93.7
JL: triage	62.5	55.7	75.6	80.0	90.0
JL: moderation	67.8	48.4	89.0	77.5	92.7
JL: routing	64.2	51.1	61.3	83.7	88.8
JL: claims	90.0	80.0	61.7	98.7	100.0
JL: reviews	70.7	42.3	83.0	87.0	87.7
JL: guard	60.6	33.2	88.0	83.2	93.8
JL: multilingual	57.8	62.5	94.5	98.4	100.0
clinc150_oos	55.7	64.8	55.2	67.8	89.5
JL: needle	50.2	47.4	77.9	87.8	95.2

Table 5: All 28 main suites, in fixed taxonomy order. Entries are percentage reference agreement; synthetic/authored sources retain their respective reference semantics. These rows are not independent benchmarks and are not averaged. Failure cases remain in denominators. Full per-source cluster intervals, type/language slices and provenance are provided by the machine-readable artifact.

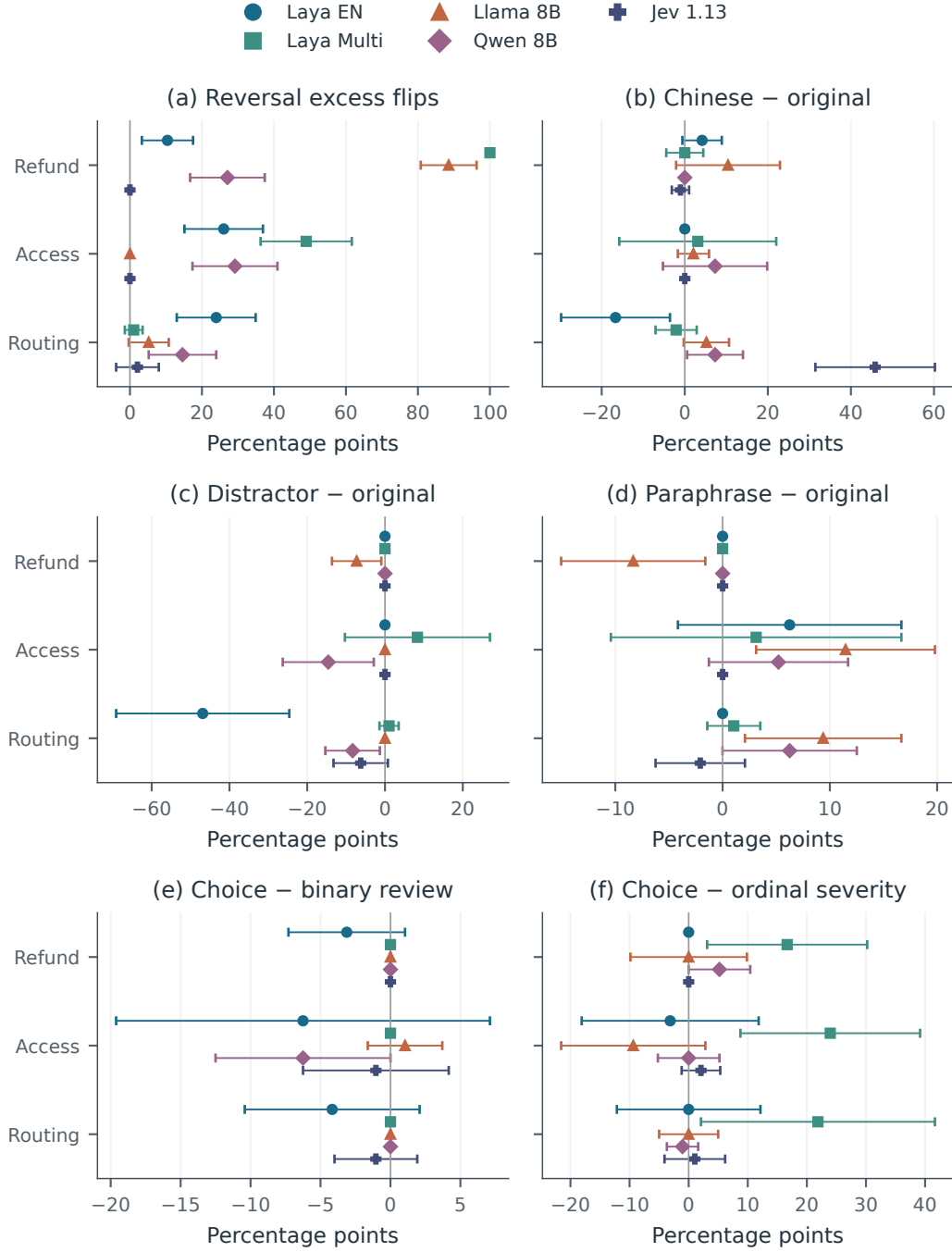


Figure 13: All six predeclared fresh-policy effects, with 96 base clusters per policy. Whiskers are simultaneous 95% intervals within the six-effect family of each model/policy, using 10,000 bootstrap draws stratified by generator seed. They do not provide paper-wide coverage. Panel (a) measures excess action flips; (b)–(f) measure correctness differences for the stated primitive. Zero-variance intervals are descriptive, not proof of invariance. Programmatic references follow visible policies. Chinese policy/question/criteria text is authored; structural field and label IDs remain in English.

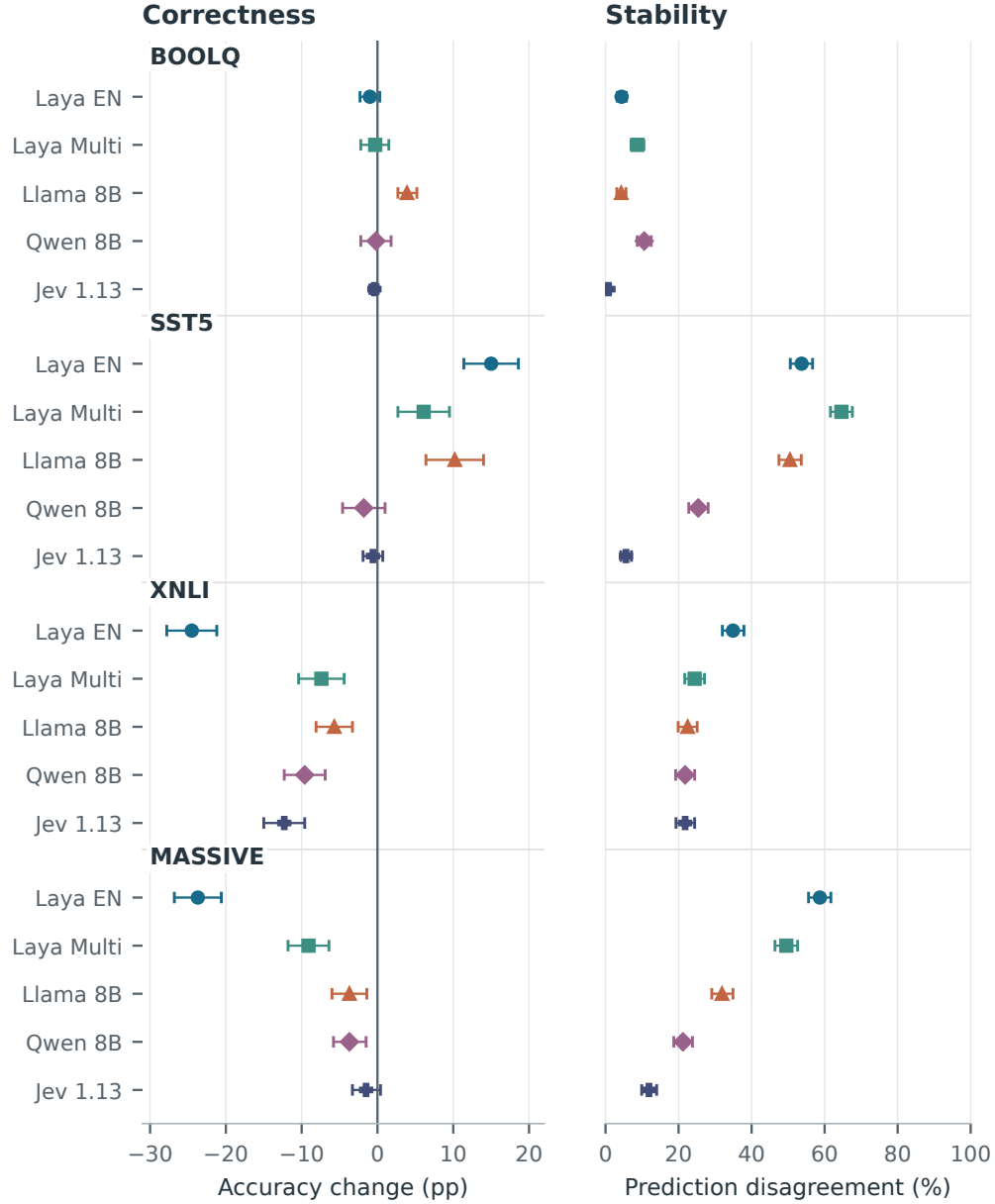


Figure 14: Exploratory paired interface contrasts against dataset-provided references on 1,000 aligned states per task: BoolQ choice minus noul, SST5 choice minus score, and Chinese minus English for XNLI and MASSIVE. Left: accuracy change; right: canonicalized prediction disagreement. Whiskers are pointwise 95% paired state-cluster bootstrap intervals (10,000 draws). Canonical label mappings are recorded explicitly; no marginal-CI subtraction is used.

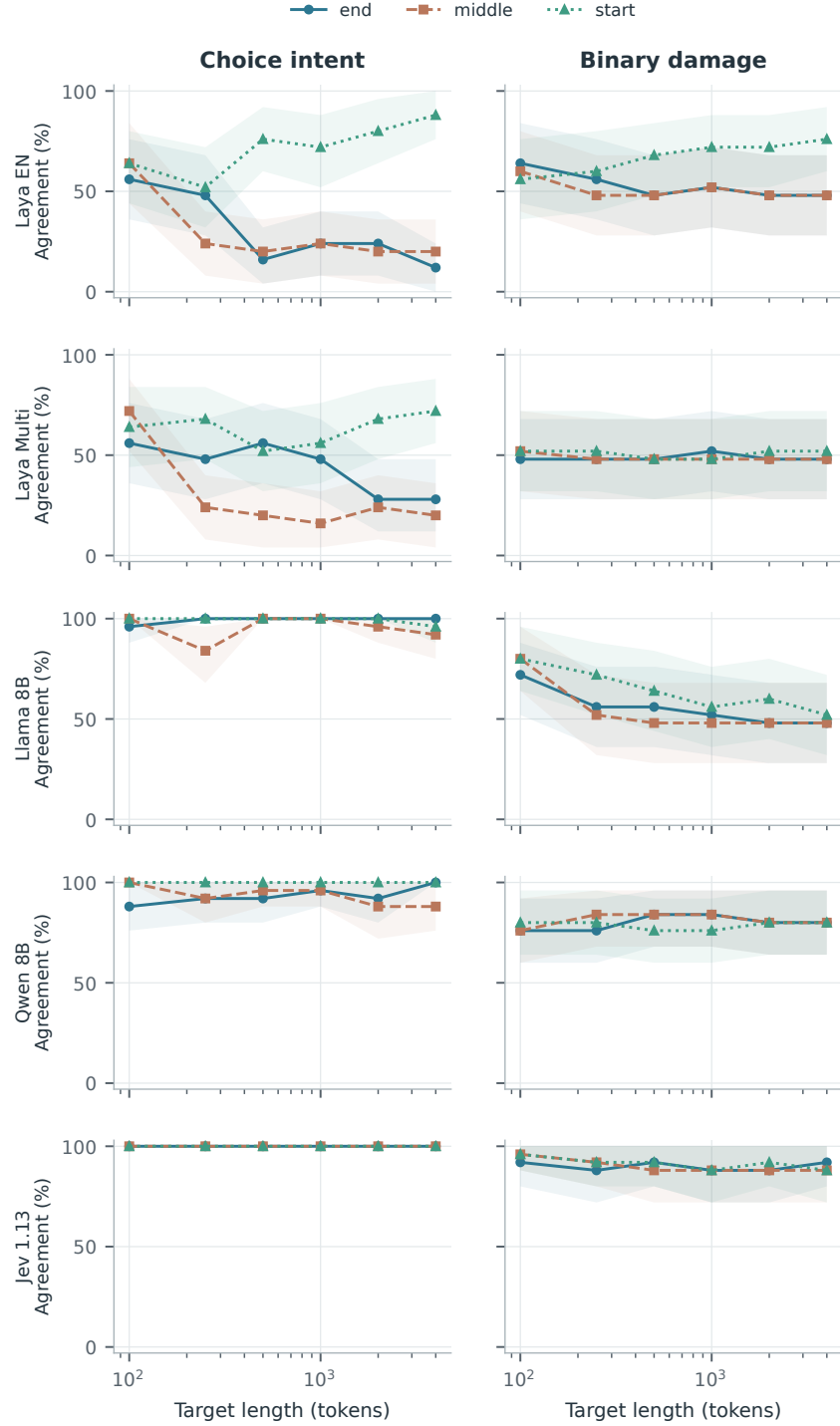


Figure 15: Programmatic needle reference agreement by source target length, insertion position and question. Rows identify models; left/right columns are the choice intent and binary damage questions. Each point uses 25 original needle clusters; shaded pointwise 95% cluster intervals resample those originals. Source length targets are not identical tokenizer token counts. The two primitives need not share the same context trend.

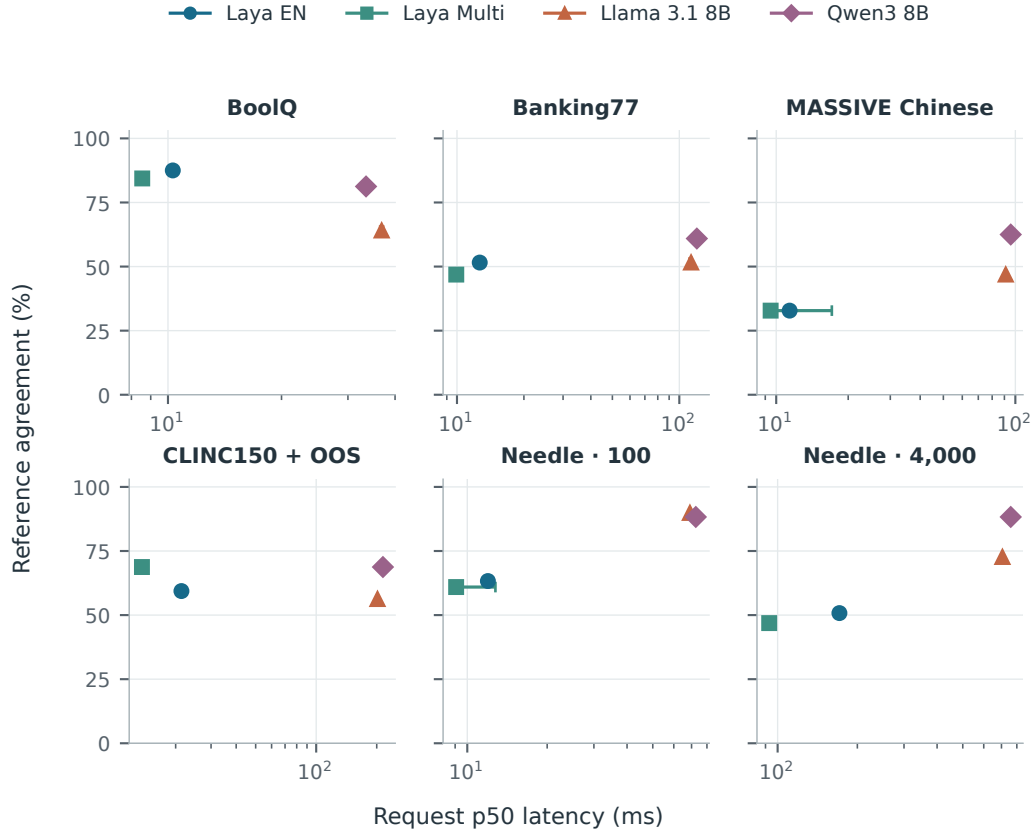


Figure 16: Resident-GPU quality/latency observations on six named workloads. Each stratum has the same 64 measured states across local models. BoolQ, Banking77, MASSIVE and CLINC150 use dataset references; the two needle workloads use programmatic references. Horizontal bars are 95% bootstrap intervals for the median of eight round p50 latencies; vertical coordinates are median reference agreement over repeated fixed states, not a new population-accuracy estimate. Jev’s hosted latency is excluded because its measurement boundary differs. No cross-workload frontier is drawn.

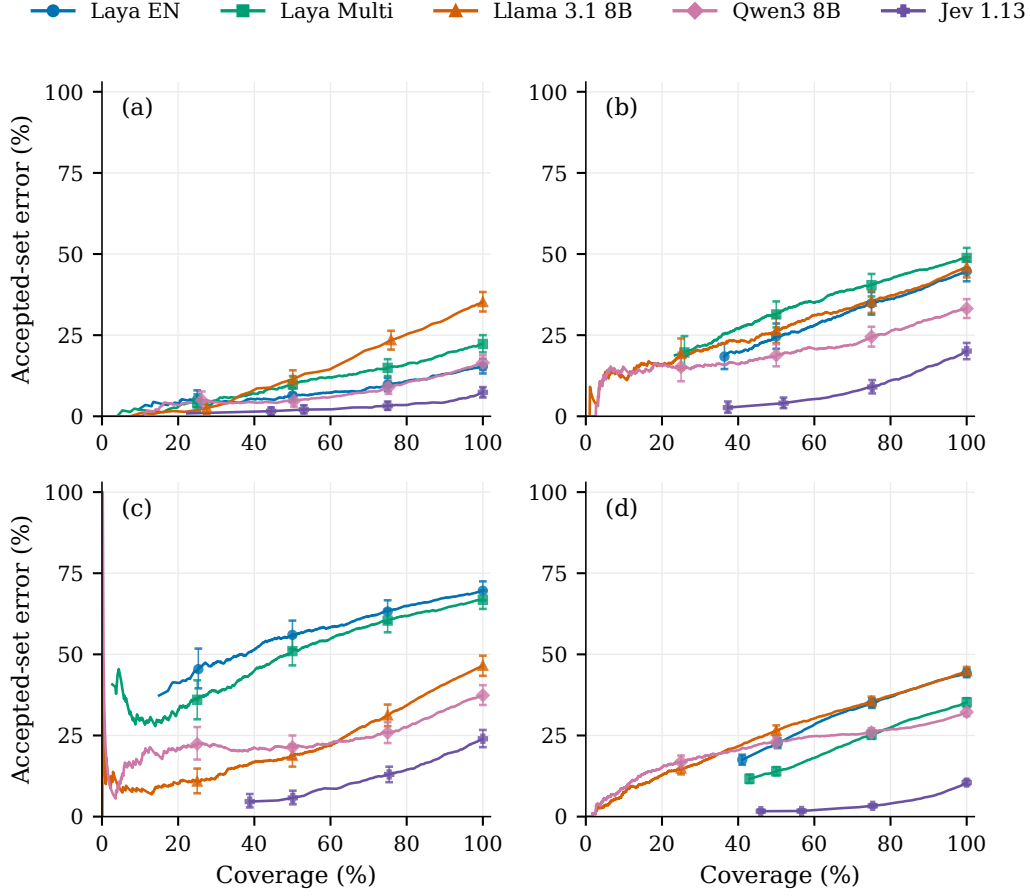


Figure 17: Maximum-probability selective error on (a) BoolQ, (b) Banking77, (c) MASSIVE Chinese and (d) CLINC150 with OOS (1,000, 1,000, 1,000 and 5,500 states). Ties are accepted together. Markers identify realized coverage at nominal 25/50/75/100% landmarks; intervals resample accepted state clusters conditional on the displayed threshold. Native and candidate-code probability semantics remain distinct. Curves describe the observed test set and do not validate a deployment threshold.

B Natural-domain sampling and reference validity

The separate legal and scientific diagnostic is deliberately balanced for paired sensitivity analysis, not a claim of natural prevalence. ContractNLI has 123 test contracts and 2,091 annotated document–hypothesis decisions, of which 220 are contradictions. The 16,000-character cap removes 24 contracts, disproportionately SEC-HTML files (13/23), versus 7/76 PDFs and 4/24 SEC-text files. The frozen 144 pairs span 91 contracts and contain 48 examples of each of the three labels; 43 of its 48 contradiction cases use four of the 17 fixed hypothesis IDs. To quantify that prior without test-label training, we compute the modal label for each hypothesis on the official 423-contract training split and apply the resulting ID-only lookup to the frozen sample: $82/144 = 56.9\%$ reference agreement (descriptive 95% contract-cluster interval [49.3, 64.3]). The same lookup reaches $1,379/2,091 = 65.9\%$ on the full official test set, illustrating how the frozen class balance changes the question. This supervised label prior is not a fair zero-shot model baseline and does not prove any system uses the shortcut. Neither the cap nor the balanced label mixture supports a prevalence-weighted comparison with ContractNLI’s published full-test scores. The observed correctness and reversal effects apply to complete selected contracts and their specified hypotheses only.

SciFact’s cited-positive subset contains 120 claim–abstract pairs from 120 distinct claims but only 111 distinct abstracts. It samples 60/64 eligible contradiction claims and 60/124 eligible support claims, so labels are balanced but claim creation mechanisms and source-paper repetition are not. The 112 development-set claims without positive evidence are excluded; the experiment neither assigns them a no-information target nor measures retrieval. Claim-cluster intervals in Figure 6 account for claim identity, not shared-abstract dependence. Re-clustering on the 111 abstracts gives nearly identical descriptive intervals: Qwen’s reversal accuracy change is -6.7 points with claim-cluster interval $[-12.5, -0.8]$ and abstract-cluster interval $[-12.7, -0.8]$; all five alternatives are recorded in the companion analysis. The supplied title and full abstract are visible, while evidence sentence indices and references are not. No local full-input audit can verify hosted server tokenization.

The first hosted attempt at this frozen experiment stopped after 216 requests when one HTTP-200 response selected a different option from its own maximum-probability option. Its append-only journal is retained as a separate aborted run. The primary Jev record is a fresh 792-request execution with an explicitly versioned runner that records such invalid typed outputs without silently selecting an alternative; no invalid decision occurred in that complete execution. This deviation is not used to replace a failed pair in the primary run. Source-text redistribution is separately governed by ContractNLI and SciFact/S2ORC notices in `research/domain_expansion_v1/DATA_LICENSES.md`; the local source-containing freeze is not shipped in Git.

Three-class SciFact source and input audit. We reconstructed the labeled development claims and complete abstracts from the official SciFact archive, binding archive, source-file, preparer, frozen-request and raw-output SHA-256 hashes in the companion artifact. Among 300 claims and 5,183 corpus abstracts, 340 citation-list occurrences reduce to 339 unique claim–document pairs because claim 1245 repeats one cited ID. The resulting pair counts are 138 SUPPORT, 71 CONTRADICT and 130 derived NOINFO; the latter require explicit membership in `cited_doc_ids` and absence of an evidence entry. Every cited ID exists in the corpus, every evidence ID belongs to its claim’s citation list, and we found no empty evidence lists, contradictory rationale labels, empty selected texts, or out-of-range rationale indices. Thirteen claims have evidence for one cited document and no evidence for another; these are different pairs, not conflicting labels. The selected 60/60/60 pairs have both unique claim and document IDs, so the 10,000-draw paired, label-stratified cluster bootstrap resamples the same 180 units across conditions. A separate source validator recomputes every selected reference and 720 request hashes; an independent analysis checker recomputes all raw counts and confusion cells. All four local adapters consumed all 720 requests without tokenizer truncation, and no system returned an invalid choice. The hosted server’s own tokenization remains unverified. SciFact claims/annotations are CC BY 4.0 and the S2ORC abstract corpus is ODC-By 1.0 under the upstream license notice; reconstructed upstream text is Git-ignored rather than redistributed.

B.1 SciFact3 codebook audit and decoder sensitivity

The exploratory four-cell diagnostic in Section 9 uses exactly the 180 previously selected, class-balanced SciFact3 claim–cited-abstract pairs. There is one pair per source claim and cited document. The observed base and joint-reversal records were reused by immutable file hashes. Display-only and

code-only requests were frozen before those missing cells were run; there was no SciFact3 factorial pilot or amendment. Before new inference, every reused prompt matched its original token sequence and length under the pinned tokenizer; checkpoint-file, chat-template, adapter, case/request, and batch-size-four hashes matched. The only model-facing user payload fields are `state`, `question_type`, `instructions`, and `candidates`. The state necessarily contains claim, paper title, and cited abstract, but not reference labels, source split, group, or source identifiers.

Model	Decoder	Base	Display only	Code only	Both
Llama 3.1 8B	native	78 (—)	61 (51)	71 (85)	69 (42)
	common tie	78 (—)	61 (51)	71 (85)	68 (46)
Qwen3 8B	native	104 (—)	114 (30)	114 (18)	101 (60)
	common tie	104 (—)	114 (30)	114 (18)	101 (60)

Table 6: Correct predictions out of 180; parentheses show semantic prediction flips relative to base. Native retains the original historical decoder, whereas common tie applies the fixed SUPPORT-CONTRADICT-NOINFO semantic priority to all four cells. Invalid counts are zero in every cell.

Exact maximum-logit ties in base/display/code/both number 6/1/11/6 for Llama and 1/6/1/0 for Qwen. Relative to native decoding, common tie changes only six Llama joint-reversal decisions; the other seven model-cell combinations are unchanged. Repeating the original base execution produced zero prediction flips for either model. An independent standard-library replay verified the SHA-256 of all eight original/new gzip result files, reconstructed all four message and candidate-assignment hashes per case, and recovered every native/common accuracy count and full base-to-cell transition matrix directly from raw stored probabilities. It also checked all reference and source-group associations without emitting source text.

Qwen3 reference class	Base	Display only	Code only	Both
SUPPORT	45/60	48/60	50/60	53/60
CONTRADICT	5/60	18/60	12/60	18/60
NOINFO	54/60	48/60	52/60	30/60

Table 7: Qwen3 correct counts by operational reference label under the common tie decoder (identical here to native decoding). The NOINFO reference means no annotated evidence in the selected cited abstract, not evidence of absence in the full paper or literature.

For the descriptive accuracy interaction (both-code) - (display-base), the Qwen estimate is -12.8 percentage points with a pointwise 95% percentile interval [-19.4, -6.1]; Llama under common tie is +7.8 points [-1.1, +16.7]. Both intervals use 10,000 paired bootstrap draws with seed 20260930, re-sampling source claims and retaining all four matched decisions together. Because each selected claim contributes only one pair, claim-cluster and case resampling coincide here. These are exploratory post-hoc-to-the-existing-outcomes, unadjusted, selected-sample intervals; they are neither population-prevalence estimates nor neural-mechanism identification. The full protocol, frozen ID/hash manifest, original raw results and replay code are under `research/scifact3_codebook_v1/`; no upstream abstract text is duplicated in its freeze.

C Reproduction and data accounting

The historical accuracy record contains 22,934 logical requests and 26,450 decisions per model, including matched controls. The timing record contains 67,584 measured request executions and 98,304 decisions across repetitions. The fresh policy diagnostic contains 288 base states, 2,304 request variants and 6,912 decisions per evaluated model. The held-out wording test adds 96 new base states and 384 requests per model. These totals refer to different collections and replication units and should not be combined into an independent sample size.

The repository retains source/revision manifests, prepared-input fingerprints, model metadata, raw per-question probabilities and predictions, input audits, and complete timing/monitor records. The paired-analysis script checks raw-file fingerprints and reference alignment before deriving effects. Figure scripts consume its JSON output or the verified performance summary, producing vector

PDF/SVG and PNG previews. The hosted runner reads credentials outside the repository, records every response attempt without authorization headers, pins the returned model ID, and resumes only under an unchanged contract.

Accuracy and typed-response validation can be recomputed on CPU from the retained raw outputs, subject to an approved artifact-access and source-license plan; no public release is claimed. New model inference additionally requires the pinned checkpoints or a provider account. Exact remote replication depends on the provider honoring its version identifier. Generator code and full programmatic confirmation cases are reproducible without downloading third-party state text.

Hosted execution deviation. The frozen retry description referred to 5xx errors generally, while the implementation enumerated HTTP 500, 502, 503, 504 and 529 (plus 429 and transport exceptions). A single HTTP 520 in the fresh refund Chinese condition received one attempt and caused three failed decisions, retained in primary denominators. Its Chinese-minus-original action effect is $-1/96$; a valid-pair sensitivity excludes that paired base state and yields $0/95$. This service failure does not support a language-regression claim. The historical matrix additionally contains four choice/probability-consistency failures affecting eight decisions, also retained. No failed answer was repaired or replayed.

One historical resume boundary also violates the protocol's global manifest-order statement: two adjacent journal entries cross that ordering. The journal remains append-only, and keyed reconciliation verifies complete request identities, payloads and outputs. No scoring or selection depends on journal position; the original bytes and the execution deviation are preserved.